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BANK OF THAILAND

MONETARY POLICY REPORT

September 2018

Monetary Policy Report

The *Monetary Policy Report* is prepared quarterly by staff of the Bank of Thailand with the approval of the Monetary Policy Committee (MPC). It serves two purposes: (1) to communicate to the public the MPC's consideration and rationales for the conduct of monetary policy, and (2) to present the latest set of economic and inflation forecasts, based on which the monetary policy decisions were made.

The Monetary Policy Committee
September 2018

Mr. Veerathai Santiprabhob	Chairman
Mr. Mathee Supapongse	Vice Chairman
Mr. Paiboon Kittisrikangwan	Member
Mr. Sethaput Suthiwart-Narueput	Member
Mr. Kanit Sangsubhan	Member
Mr. Subhak Siwaraksa	Member
Mr. Somchai Jitsuchon	Member

Monetary Policy in Thailand

Monetary Policy Committee

Under the Bank of Thailand Act, the Monetary Policy Committee (MPC) comprises the governor and two deputy governors, as well as four distinguished external members representing various sectors of the economy, with the aim of ensuring that monetary policy decisions are effective and transparent.

Monetary Policy Objective

The MPC sets monetary policy to promote the objective of supporting sustainable and full potential economic growth, without causing inflationary problems or economic and financial imbalances or bubbles.

Monetary Policy Target

The Cabinet approved the annual average headline inflation target of 2.5 $\pm$ 1.5 percent as the target for the medium term and for 2018. The inflation target is to assure the general public that the MPC will take necessary policy actions to return headline inflation to the target within an appropriate time horizon without jeopardizing growth and macro-financial stability. In the event that headline inflation deviates from the target, the MPC shall explain the reasons behind the target breach to the Minister of Finance and the public, together with measures taken and estimated time to bring inflation back to the target.

Monetary Policy Instrument

The MPC utilizes the 1-day bilateral repurchase transaction rate as the policy interest rate to signal the monetary policy stance.

Evaluation of Economic Conditions and Forecasts

The Bank of Thailand takes into account information from all sources, the macroeconomic model, data from each economic sector, as well as surveys of large enterprises, together with small and medium-sized enterprises from all over the country, and various financial institutions to ensure that economic evaluations and forecasts are accurate and cover all aspects, both at the macro and micro levels.

Monetary Policy Communication

Recognizing the importance of monetary policy communication to the public, the MPC employs various channels of communication, both in Thai and English, such as (1) organizing a press statement at 14:00 on the day of the Committee meeting, (2) publishing edited minutes of the MPC meeting two weeks after the meeting, and (3) publishing the *Monetary Policy Report* every quarter.

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Executive Summary

Monetary Policy Conduct in the Third Quarter of 2018

The Committee assessed that the Thai economy continued to gain traction but was still subject to significant risks from the U.S. trade protectionism measures and retaliatory measures by its trading partners. Headline inflation was projected to trend up, although there were downside risks with respect to highly volatile fresh food prices. Financial stability remained sound but there was a need to monitor pockets of risks that might pose further vulnerabilities to financial stability in the future. **The Committee weighted various factors in determining the most appropriate course of monetary policy and voted 6 to 1 and 5 to 2 to maintain the policy rate at 1.50 percent on meetings on August 8 and September 19, 2018 respectively.** In deliberating their decision, the Committee viewed that the current accommodative monetary policy stance remained necessary to help sustain economic growth and foster headline inflation to move within the target. Under the Committee's assessment, the policy rate at 1.50 percent would facilitate sufficiently accommodative financial conditions as reflected in the new loan rate (NLR) which remained at a low level. Despite some increases, the real policy rate and government bond yields remained accommodative overall and continued to support business financing. Meanwhile, financial stability risks remained manageable. Nonetheless, some Committee members voted to raise the policy rate by 0.25 percentage point to 1.75 percent as the economy was sufficiently robust and expanded above potential. In their view, the financial system showed signs of increased vulnerabilities at a broader scale as prolonged accommodative financial conditions led consumers and businesses to underprice risks. Thus, some members voted to raise the policy rate to curb the build-up of vulnerabilities to financial stability and also to start building up policy space for the future.

The Committee viewed that the current accommodative monetary policy stance remained necessary. Members also discussed conditions and appropriate timing to begin normalizing monetary policy in the future. Under the Committee's view, should economic expansion continue and inflation move more firmly within the target, the need for currently extra accommodative monetary policy would start to be gradually reduced, and the need for a policy rate increase in order to build up policy space in the future would be increasing. The Committee's evaluation of the appropriate conditions would be data dependent, including careful assessment of the outlook of economic growth and inflation, as well as risks especially on the external front.

Assessment of the Economic and Financial Outlook as the Basis for Policy Formulation

1. Global Economy

The global economy was projected to continue growing in line with the previous assessment despite a slight slowdown in 2019. Thailand's trading partner economies continued expanding in 2018. In particular, the U.S. economy recorded robust growth due to strong economic fundamentals and fiscal stimulus. Meanwhile, growth of the euro area and Asian economies slightly slowed down. Growth of trading partner economies would be lower somewhat in 2019 owing to the impact of trade protectionism measures on global trade volume, some effects of the 200 billion dollar worth of tariff that the U.S. imposed on China's exports, and tightening financial conditions in several countries. **The Committee thus maintained the growth forecast for Thailand's trading partners at 3.8 percent in 2018, while revising down the growth forecast from 3.6 to 3.5 percent in 2019.** Risks that would warrant monitoring included the U.S. trade protectionism measures which could intensify and lead to retaliatory measures. Moreover, geopolitical risks remained uncertain and could escalate to impact financial and commodity markets as well as the real sector. In addition, concerns over China's financial stability would continue to warrant monitoring despite regulatory improvements by the Chinese authorities.

Most central banks maintained accommodative monetary policy stance, while some central banks in the region raised their policy rates. The Bank of Japan was expected to keep their short- and long-term target rates on hold for some period after adjusting their forward guidance and allowing more flexibility in movements of the 10-year government bond yield. The European Central Bank would likely maintain its policy rate until the second half of 2019. Meanwhile, the U.S. Federal Reserve would continue its monetary policy normalization. The Bangko Sentral ng Pilipinas raised the policy rate to stabilize inflation, while Bank Indonesia hiked its policy rate to curb

volatility in financial market. Going forward, continued economic growth and rising inflation toward target would facilitate monetary policy normalization for other central banks in the region.

Emerging markets (EMs) experienced capital outflows owing to concerns over intensifying trade protectionism measures, higher costs of financing in global financial markets due to the increases in the U.S. policy rate, and weak economic fundamentals of some EMs. As a consequence, foreign investors sold assets in vulnerable EMs such as Turkey, Argentina, and South Africa and redirected their investment toward markets with stronger external stability such as Thailand, South Korea, and Taiwan. However, global financial markets would likely remain volatile and warrant close monitoring in the period ahead.

2. Financial Conditions and Financial Stability

Thailand's financial conditions remained accommodative. Short-term Thai government bond yields increased but remained below the policy rate. Medium-term bond yields rose due to domestic factors as the latest economic outturns were better than market expectations. Meanwhile, long-term bond yields increased only slightly due to higher demand for long-term bonds particularly from foreign investors. The new loan rate remained at a low level. Private credit expanded for both business and household sectors. Businesses continued to seek funding through both debt and equity instruments. The Thai baht appreciated against the U.S. dollar from the previous quarter following better-than-expected economic outturns, greater clarity on the timeline of the upcoming general election in Thailand, and the investment of foreign investors in EMs with strong external stability. The real effective exchange rate (REER) appreciated.

Financial stability remained sound but there remained pockets of risks that warranted monitoring. These included, first, increased vulnerability in the property sector. As financial institutions competed in extending mortgage loans and became willing to bear higher risks, credit standards became looser. Furthermore, the share of non-performing loans in mortgage loans increased. Meanwhile, the oversupply of condominiums in certain areas remained high. Second, elevated household debt had yet to show clear signs of deleveraging, while debt serviceability of households and small businesses deteriorated. Third, the search-for-yield behavior persisted in the prolonged low interest rate environment which could exacerbate underpricing of risks by the private sector. For instance, saving cooperatives continued to provide high returns to members resulting in high growth of their assets, which in turn could pressure them to search for higher returns. In addition, in the prolonged low interest rate environment, the issuance of corporate bonds was concentrated among large corporations, which tended to invest more in non-core businesses and overseas enterprises. This would pose greater risks to business operations.

3. Economic and Inflation Outlook

The Thai economy was projected to record robust and continued growth at 4.4 and 4.2 percent in 2018 and 2019, respectively. The growth forecast was consistent with the assessment in the previous *Monetary Policy Report*. Key economic drivers stemmed from improvements in private spending, both through consumption and investment, thanks to a more broad-based increase in employment and support from greater clarity on public investment projects, as well as continued expansion of merchandise exports and tourism. Nonetheless, public spending growth was projected to be lower than expected.

Merchandise exports were projected to grow in line with global demand and partly benefited from the relocation of production base to Thailand for some export industries. The value of merchandise exports in 2018 was projected to expand at 9.0 percent unchanged from the previous assessment. However, export growth in 2019 was expected to slow down to 4.3 percent from the previous estimate of 5.0 percent given the assessment of some impacts of the U.S. trade protectionism measures and China's retaliatory measures on global trade volume and Thailand's trading partner economies. Meanwhile, the Committee assessed that trade protectionism measures could intensify and rapidly develop which would put pressures on international trade and investment. This would in turn affect both directly and indirectly Thailand's exports. Thus, the Committee would monitor more closely developments of trade policies and negotiations, their effects on supply chains, and impacts on Thai businesses.

Exports of services in 2019 were expected to rise because some Chinese tourists postponed their travel plans from the latter half of 2018 following the Phuket tour boat sinking incident in early July. The projected number of foreign tourists in 2018 was kept at 38.3 million unchanged from the previous assessment due to a higher-than-expected outturn in the second quarter, which offset a drop in the number of Chinese tourists during the second half of 2018. The impact of the Phuket boat incident, nonetheless, was expected to be short-lived given signs of recovery in Chinese tourist figures in many areas. For 2019, the projected number of foreign tourists was revised up from previously assessed to 40.6 million due to postponed travel plans by some Chinese tourists from 2018, new flight routes, and greater airport capacity following increased management efficiency.

Private consumption was expected to achieve higher growth supported by improvements in household income. Medium- and high-income households in the non-agricultural sector experienced steady income growth. Low-income households also experienced income growth supported by improvements in employment in most sectors. Meanwhile, farm income expanded on the back of higher agricultural production and support from government policies. However, elevated household debt would cause households to allocate part of their income for debt repayment. Moreover, structural changes in the labor market such as adoption of automation in place of human labor in the production process limited wage increases. As a result, purchasing power would recover gradually.

The government would help drive the economy despite some slowdown in public spending compared with the previous assessment. Government consumption expenditure, particularly compensation of civil servants with regard to salary and medical expenditure, was projected to decrease given the policy to replace vacant job positions with contract workers. Public investment decreased for both the central government and state-owned enterprises. For the central government, investment were revised down due to lower efficiency in budget disbursement by some government units, following construction problems that included limited construction capacity, land reclamation issues, and larger-than-expected impacts of the Public Procurement and Supplies Management Act, B.E. 2560. For state-owned enterprises, some investment projects faced operational difficulties including project revisions, funding reviews, and bidding process delays.

Private investment was projected to gain traction with further improvement in 2019 thanks to better-than-expected private consumption, together with capital outlays following production relocation to Thailand of some export-oriented industries during late 2018 and 2019. Moreover, other supporting factors included (1) higher capacity utilization in various industries such as automobiles, electronics, and chemical products, (2) greater clarity on investment plans of large companies, (3) higher demand for corporate credits, and (4) improved investment sentiment following greater clarity on infrastructure investment projects, the Eastern Economic Corridor (EEC), and public-private partnership (PPP).

The outlook for inflation in 2019 was expected to rise in line with the previous assessment. Price increases in the fresh food items was projected to be slower than expected mainly because supplies of meat, vegetables, and fruits were expected to increase more than previously assessed due to favorable weather conditions, the government's irrigation management that gave priority to agricultural purposes, and advancement in agricultural technology. Meanwhile, energy prices rose in tandem with global crude oil prices. Demand-pull inflationary pressures lowered partly due to structural factors including improvement in production technology that resulted in lower costs of goods and services, expansion of e-commerce, and globalization that led to intense competition and difficulty in raising prices. Such factors could result in more persistent inflation than in the past, although the economy expanded in line with its potential. **The Committee therefore projected headline inflation to average at 1.1 percent in 2018 and 2019 and core inflation to average at 0.7 and 0.8 percent in 2018 and 2019, respectively.**

Risks to the growth projection were expected to tilt downward but with a smaller degree than the previous assessment due to an increased possibility that the Thai economy would outperform the baseline projection supported by (1) domestic demand growth that could be higher than expected thanks to infrastructure investment and government stimulus measures to support private spending that could be additionally announced, (2) growth of Thailand's trading partner economies that could be higher than expected on account of continued improvements in the U.S. economy with support from tax reforms, (3) a Chinese economic slowdown that could be less severe than expected if the Chinese government were to announce additional stimulus measures, which would eventually

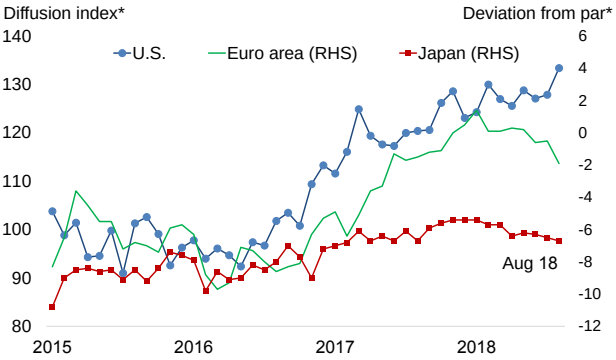
lead to better-than-expected Asian exports, and (4) the number of Chinese tourists that could be larger than expected following the recovery after the Phuket boat incident. However, there remained possibility that the Thai economy would grow at a rate below the baseline projection despite the baseline projection already taking into account some impact of the 200 billion dollar worth of tariff that the U.S. imposed on China's exports. Such possibility is due to uncertainties regarding (1) the U.S. trade protectionism measures and additional retaliatory measures from major economies as well as intensifying competition resulted from trade diversion which could weigh on Thailand's exports and investment, (2) Thailand's trading partner economies which could expand at lower rates than expected due to tensions arising from geopolitical risks and economic problems among EMs, (3) domestic purchasing power that improved only gradually could affect private consumption growth, and (4) the Public Procurement and Supplies Management Act which could delay budget disbursements for some government agencies as well as public investment projects by state-owned transportation enterprises which might experience delays due to reviews of investment project approvals. **Meanwhile, risks to the forecasts of headline and core inflation were expected to tilt downward** in line with risks to the growth projections and highly volatile fresh food prices.

1. Global Economy

Major advanced economies continued to expand thanks to strong economic fundamentals. Nonetheless, growth momentum would slow down somewhat mainly owing to the impact of trade protectionism measures between the U.S. and China.

The U.S. economy was expected to exhibit continued robust growth due to strong economic fundamentals. In the second quarter of 2017, the economy was growing at a faster-than-expected pace as private consumption rebounded after a temporary slowdown in the previous quarter due to the unusually cold weather and delayed tax refunds. Looking ahead, the U.S. economy would continue expanding with support from strong labor market conditions, policies regarding personal and business income tax cuts, robust consumer confidence, (Chart 1.1) as well as improvements in household financial positions. However, the impact from intensifying trade protectionism measures between the U.S. and China could somewhat undermine the growth outlook as some businesses might face difficulties in the short term adjustments. Although the impact on the U.S. economy overall did not yet materialize, the effects on imports of goods that were directly levied tariffs were already observed. For instance, numbers of imported washing machines declined and costs of importing metal and aluminum products rose which were partly transmitted to related downstream industries. **The Japanese economy was expected to continue expanding in line with the previous assessment, although growth momentum would slightly slow down in 2019 due to the impact from trade protectionism measures between the U.S. and China**. During the first half of 2018, Japan would likely grow in line with the previous projection. Despite a sharp slowdown during the first quarter due to domestic demand that was subjected to temporary factors such as unusually cold weather, strong economic fundamentals including robust consumer confidence, strengthening labor market, high corporate profits, and the continuation of monetary policy accommodation enabled higher-than-expected domestic demand growth in the second quarter and continued growth momentum in the remainder of the year. In 2019, growth was projected to slightly slow down on account of exports which would exhibit slower growth following global trade condition after having accelerated in 2018, coupled with the effects of trade protectionism measures on Japanese production supply chains. **The euro area was expected to slow down** mainly due to private consumption which was partly undermined by concerns of consumers and businesses over political issues in the euro area. Moreover, exports were expected to experience indirect effects of trade protectionism measures which would curb growth momentum somewhat. Nonetheless, monetary policy accommodation, continued strengthening labor markets, and rising wages in many countries including Germany and France would support the continuation of the economic growth in the period ahead.

Chart 1.1 Consumer confidence continued to remain high in the US, while there were signs of slowing down in the euro area and Japan



Note:*Euro area's par = 0 and Japan's par = 50
Source: Bloomberg

Economies of China and Asia were projected to continue expanding on the back of domestic demand growth despite tightening financial conditions. Merchandise exports would continue to rise in line with global demand. However, intensifying trade protectionism measures could weigh on economic growth in China and Asia going forward.

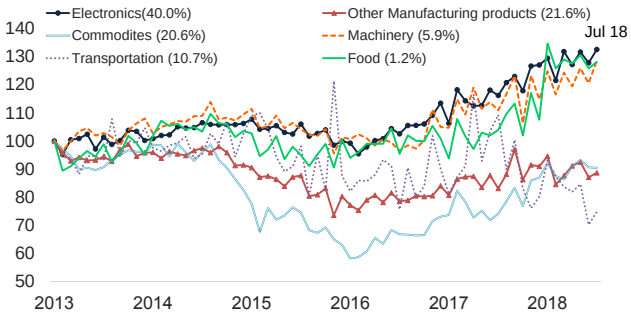
China saw a continuation of growth but could experience some slowdown due to intensifying trade protectionism measures between the U.S. and China. Domestic demand indicators in the second quarter of 2018 slowed down than expected following tightening financial condition after series of financial stability measures were released by the Chinese government to curb debt levels in the economy, particularly measures to contain shadow banking risks and tighten credit standards on credits to local government. However, economic stimulus measures that would be implemented in the period ahead were expected to partly alleviate adverse effects from financial stability measures and the U.S. trade protectionism measures. The stimulus measures included additional fiscal stimulus through accelerated infrastructure investment projects and tax cuts in order to reduce burden on businesses and households. Besides, the initial impact of the U.S and China's trade protectionism measures on Chinese exports and the overall Chinese economy were still limited. Nonetheless, intensifying trade protectionism measures would consequently affect China's growth and warrant close monitoring going forward.

Asian economies, excluding Japan and China, were expected to exhibit slower growth. In the second quarter of 2018, several economies expanded at lower-than-expected rates due to temporary factors that somewhat set back economic activities such as production bottlenecks in agricultural and mining sectors in Malaysia. However, Asian economies were expected to continue expanding in the period ahead mainly on the back of domestic demand, underpinned by strong labor markets and robust confidence among consumers and businesses. Nevertheless, tightening financial conditions would have some effects on domestic demand expansion. Meanwhile, exports would expand in line with global demand (Chart 1.2) despite some slowdown due to the following factors; (1) slower growth in global demand after accelerating in the previous year, especially for electronics, and (2) indirect effects of the U.S. and China's trade protectionism measures on global supply chain.

Chart 1.2 Asian exports continued to improve and expand across various product categories

Asian exports value* classified by product categories

Index, sa (Jan 2013 = 100)



Note: *Asian exports include Hong Kong, Taiwan, Korea, Malaysia and Singapore.
( ) share of total exports in 2017
Commodity-related products include crude oil, metals, chemicals, rubber, and vegetable oil.
Other manufacturing products include textile, papers, furniture, footwear and miscellaneous

Source: CEIC

The growth projection for Thailand's trading partner partners was slightly lower than the previous estimate. This was attributed to both temporary factors which affected growth in several countries during the first half of this year and some impact from additional trade protectionism measures between the U.S. and China. Risks to the growth projection remained unchanged and still tilted downward.

Thailand's trading partner economies were expected to gain traction despite some slowdown in 2019 in comparison to the assessment in the previous *Monetary Policy Report.* In 2018, Thailand's trading partner economies were expected to grow at rates unchanged from the previous assessment. Key contribution would be from the U.S. economy with its strong economic fundamentals and support from fiscal stimulus. Meanwhile, growth of economies in euro area and Asia (excluding China) would slow down as growth outturns were lower-than-expected during the first half of the year. In 2019, growth momentum of Thailand's trading partners would slightly decline as global trade could slow down more than expected following the impact of trade protectionism measures. In projecting the growth momentum, the Committee included some impact of trade protectionism measures between the U.S. and China including the 200 billion dollar worth of tariff that the U.S. imposed on China's goods and tightening financial conditions in some countries such as the Philippines and Indonesia. The Committee, therefore, maintained the growth projections of Thailand's trading partners at 3.8 percent in 2018 and revised downward the projection for 2019 to 3.5 percent. Going forward, strong economic fundamentals, especially domestic demand, would support global economic growth.

The Committee assessed risks to growth of Thailand's trading partners to remain broadly unchanged from the estimates in the previous *Monetary Policy Report* which still titled downward based on the following factors. First, trade protectionism measures of the U.S. could intensify and trigger retaliatory measures from several countries which would have direct impact on global trade and Thailand given supply chain interconnectedness. Moreover, indirect effects would include lower private investment growth due to increased uncertainty. Second, geopolitical risks remained uncertain including the U.S.-North Korea denuclearization agreement that had yet to reach a concrete solution in spite of negotiation developments as well as tensions in the Middle East which could affect fluctuations in the financial markets, commodity markets especially oil prices, and the real economy. Third, China's financial stability concerns still warranted monitoring despite regulatory improvements by the Chinese authorities.

Table 1.1 Assumption on trading partner growth

Annual change (%YoY)	Weight (%)	2017*	2018	2019
United States	14.9	2.2	2.8 (2.7)	2.3 (2.3)
Euro area	10	2.5	2.1 (2.2)	1.8 (2.0)
Japan	13.6	1.7	1.2 (1.2)	1.0 (1.0)
China	15.7	6.9	6.6 (6.6)	6.2 (6.3)
Asia (excluding Japan and China)**	37.4	4.6	4.3 (4.4)	4.1 (4.2)
Total***	100	3.9	3.8 (3.8)	3.5 (3.6)

Note: *Outturn
**Weighted by a share of Thailand's total exports to 7 trading partners in 2014, namely Singapore (6.5%), Hong Kong (7.9%), Malaysia (8.0%), Taiwan (2.5%), Indonesia (5.9%), South Korea (2.8%), and the Philippines (3.7%)
***Weighted by a share of exports from Thailand to 13 trading partners in 2014 (including the United Kingdom and Australia)
( ) as reported in Monetary Policy Report, June 2018

Most central banks continued to maintain accommodative monetary policy. Meanwhile, some central banks in the region raised policy rates.

Most central banks continued to maintain accommodative monetary policy. The Bank of Japan (BOJ) adjusted its forward guidance to keep short-term and long-term interest rates on hold for some period. The main objective was to support growth recovery and foster inflation to rise to the 2 percent target, as well as preempting the impact of the consumption tax increase in October 2019. In addition, the BOJ would also maintain their purchase of bonds and other assets while allowing more flexibility on movements of the 10-year government bond yields and increasing the share of exchange traded funds (ETF) in the bond purchase program. The European Central Bank (ECB) would continue its net asset purchases until December 2018 and was expected to maintain its policy rate until the second half of 2019. Meanwhile, the Federal Reserve (Fed) was expected to continue its monetary policy normalization on the back of robust economic growth and continued tightening labor market. These conditions would push inflation to move in line with the target. Under the Committee's assessment, the Fed was expected to raise the federal funds rate in total of four times this year and three times in 2019, in line with the previous forecast.

Some Asian central banks raised their policy rates. The Bangko Sentral ng Pilipinas raised the policy rate in August 2018, after hiking the rate twice in the second quarter to stabilize acceleration in inflation which exceeded the 2-4 percent target. This was due to the effects from tax reforms and rising oil prices. Bank Indonesia also increased its policy rate in August 2018 after hiking rate three times during the second quarter to curb pressures on capital outflows and currency depreciation. In addition, other central banks started to signal changes in monetary policy directions given continued economic growth and rising inflation toward target which would facilitate monetary policy normalization in the period ahead.

Emerging markets (EMs) experienced capital outflows owing to concerns over intensifying trade protectionism measures and higher costs of financing in the global financial market due to the increases in the U.S. policy rate. Moreover, fundamental economic problems in some EMs led foreign investors to sell assets.

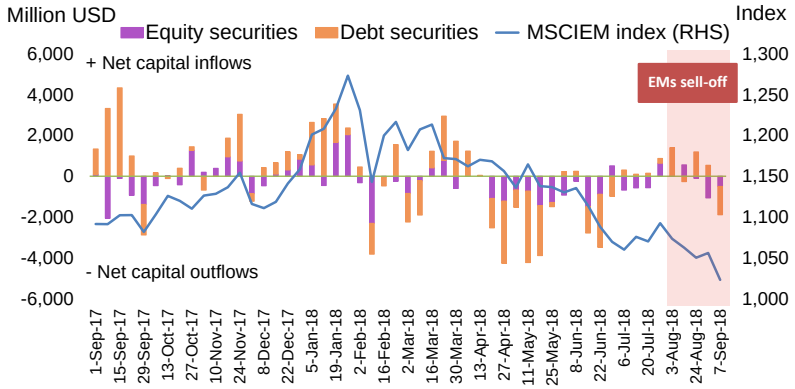
At the end of the third quarter of 2018, EMs with weak economic fundamentals including Turkey, Argentina, and South Africa experienced capital outflows from both bond and equity markets. Such vulnerabilities included large current account deficit, persistent fiscal deficit, accelerated inflation, and growth contraction amid concerns over intensifying trade protectionism policies and higher costs of financing in the global financial market following the U.S. policy rate. Moreover, specific domestic factors in some countries also posed concerns such as political tensions between the U.S. and Turkey. These factors exerted downward pressures on exchange rates of EMs and equity prices as reflected in the MSCI EM[1/] which plunged 20 percent from the highest point in January 2018 (Chart 1.3). However, investors returned to the EMs after worries over trade protectionism measures moderated. Investors redirected their investment from vulnerable EMs to those with stronger external stability such as Thailand, South Korea and Taiwan.

[1/] The MSCI Emerging Markets (MSCI EM) is an index that reflects movements of medium- and large-sized equity prices in 24 emerging markets, calculated by the MSCI.

Looking ahead, global financial markets would experience high volatilities. International capital flows between EMs would fluctuate given the Fed's rate hike, intensifying trade protectionism measures, and investors' concerns over risks of wider contagion to other vulnerable EMs. Such issues would warrant close monitoring.

Chart 1.3 Foreign investors sold EM assets following concerns over intensifying trade protectionism measures, higher costs of funding in global financial markets, and the economic vulnerabilities of some EM countries.

Net capital inflows to EM* (weekly) and MSCIEM index



Note: *EM includes Thailand, Indonesia, India, South Africa and Turkey
Sources: Bloomberg and Institutional Institute of Finance

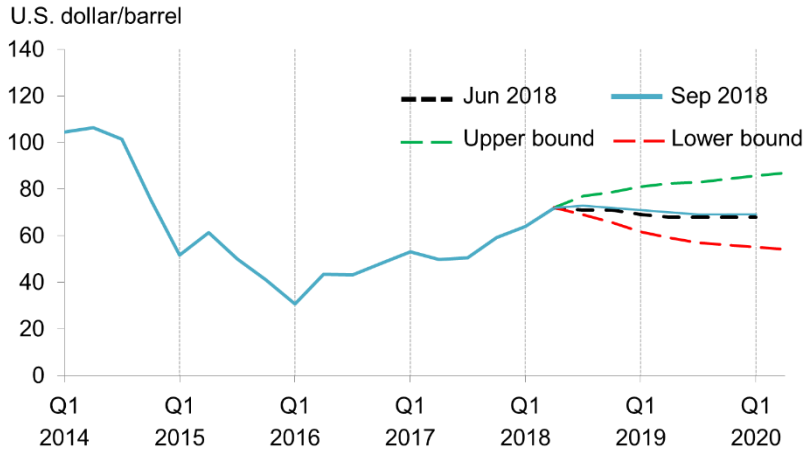
Crude oil prices continued to increase in the third quarter of 2018 due to higher demand. Moreover, investors were concerned about the adequacy of oil supply in the short run as the reduction in supply might be more than expected following the situations in Iran and Venezuela. In the period ahead, crude oil prices were projected to decline as supply from U.S. shale oil producers would gradually increase.

In the third quarter of 2018, the Dubai crude oil prices continued rising from the previous quarter. The increase was mainly due to tightening market conditions as demand for oil continued to rise in line with global economic growth. Meanwhile, supply of oil would decline more than market expected. The decrease in supply was attributed to, first, Iran's declining production after the U.S. was pulled out of the U.S.-Iran Joint Comprehensive Plan of Action (JCPOA) and imposed a sanction on Iran. Second, Venezuela's oil production declined as economic downturns weighed on the potential of oil companies regarding capital and workers as well as the damage from the ship crash incident at Venezuela's oil port. Going forward, however, crude oil prices were expected to trend down as crude oil supply from U.S. shale oil producers would gradually increase and oil producers could increase their productions in order to stabilize global oil prices. However, oil prices would not fall by a large extent as they would still be shored up by steadily growing demand following expanded global economy.

The Committee slightly revised up the projection for Dubai crude oil prices throughout the forecast horizon as crude oil supply would be lower than expected. Oil prices were revised up from 69.2 to 70.3 dollars per barrel in 2018, and from 68.3 to 69.8 dollars per barrel in 2019 (Chart 1.4). **Risks to the projection were balanced.** Upside risks that could push prices above the baseline projection included geopolitical risks and Iran's oil supply that might decline more than expected following sanctions imposed by the U.S. On the

other hand, downside risks consisted of higher-than-expected U.S. shale oil production. Moreover, the impact of trade protectionism measures on global trade volume and global economic growth might be greater than anticipated.

Chart 1.4 Dubai crude oil price in 2018Q3 continued to increase from the previous quarter. This was due to rising demand and investor concerns over the adequacy of oil supply in the short run as the oil stock would be lower-than-expected following situations in Iran and Venezuela.



2 .The Thai Economy

2.1 Recent Developments

The Thai economy continued to gain further traction in the second quarter of 2018, driven by external and domestic demand.

In the second quarter of 2018, the Thai economy expanded 4.6 percent from the same period last year. The key driver was the expansion of merchandise exports across almost all major export destinations and product categories, in tandem with continued improvements in trading partner economies and global trade volume. In particular, manufactured exports that recorded robust growth were, for example, hard disk drives, computer parts, and automobiles. Exports of services continued to expand mainly on the back of increasing numbers of Chinese and ASEAN tourists, albeit at a somewhat slower rate due to the declining low base effect from government actions against illegal tour operators in 2016. The numbers of tourists from Europe and Russia also grew at a slower pace in the second quarter of 2018 due to temporary factors, namely the overlapping months of Easter holiday and the World Football Cup events. The growth momentum was also supported by improvements in domestic demand. Growth in private consumption, in particular consumption of durable goods, accelerated from the previous quarter in line with robust growth in auto hire purchase loans. Spending on other consumption goods also recorded healthy growth. The key supporting factor was stronger purchasing power of households and improved consumer confidence. Private investment continued to expand, particularly for investment in machinery and equipment such as vehicles and transport equipment. Growth in construction investment slightly slowed down. Meanwhile, public expenditure grew at a slower pace due mainly to the slowdown in procurement of goods and services and compensation of civil servants. However, public investment continued to expand on account of construction projects of state-owned enterprises such as the Metropolitan Electricity Authority, the Electricity Generating Authority of Thailand, and the State Railways of Thailand. Overall, the Thai economy in the second quarter of 2018 recorded a 1.0 percent growth from the previous quarter after seasonal adjustment, slowed down from 2.1 percent in the previous quarter.

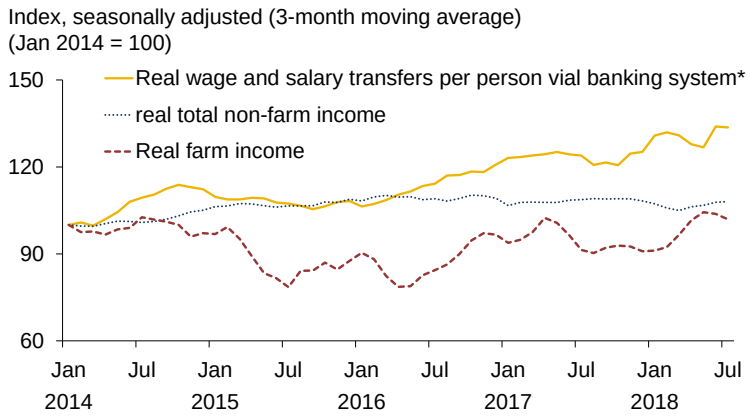
The Thai economy continued to gain further traction in **the third quarter of 2018,** as reflected by recent economic indicators, despite a somewhat slowdown in the growth of merchandise and service exports. Merchandise exports were affected by the impact of the protectionist trade policy, slower growth of Thailand's trading partner economies and global trade volume. In addition, agricultural exports contracted with declines in outputs of some products such as tapioca. Growth in exports of services slowed down in line with lower number of Chinese tourists following the Phuket tour boat sinking incident in July 2018. However, the impact was expected to be short-lived and affect only certain Chinese tourist groups. Indicators also showed a slowdown in private investment after earlier acceleration of investment in machinery and equipment. Growth in public expenditure slowed down. Growth in current expenditure slowed down for both procurement of goods and services and compensation of civil servants, while growth in capital expenditure slowed down due to a constraint of budget disbursement efficiency of public sector agencies. However, private consumption continued to record robust growth across all categories thanks to higher household income and overall improvement in consumer confidence.

The positive spillovers from the continued economic growth were extended to the labor market, with increasingly broad-based improvements in employment and income. However, purchasing power of households slowly improved owing to the impact of structural changes on wage adjustments while household debt remained high.

Purchasing power of households improved and the improvements were more broad-based thanks to the continued economic expansion (Chart 2.1). Employment and incomes of **non-farm households** improved in both the manufacturing and service sectors with signs of more positive spillover effects on low-income households. This was reflected in a gradual rise in employment among low-income workers. Meanwhile, incomes of middle- to high-income households continued to increase as indicated in higher per capita salary transfer via financial institutions. Employment of **farm households** was at a healthy level. Farm income expanded owing to the increased outputs in almost all product categories, especially rubber and rice which were grown by a majority of Thai farmers. Outputs increased thanks to favorable weather condition and water supply. Prices of some crops rose, notably jasmine rice and tapioca, on account of higher global demand and limited supplies, while rubber prices continued to decline. The recent floods affected only small agricultural areas as the main dams in the major river basins, which covered a majority of agricultural areas, were able to contain the water.

Chart 2.1 Overall purchasing power gradually improved

Various household income indicators



Note: *wage and salary transfer transactions are calculated from 2 databases: (1) Commercial banks reporting transactions to Bank of Thailand database which covers 90% of all retail transfer transations and (2) Interbank Transaction Management and Exchange (ITMX) database which covers 10% of all retail transfer transactions.

Sources: Bank of Thailand, Office of Agricultural Economics, National Statistical Office, Ministry of Commerce, calculations by Bank of Thailand

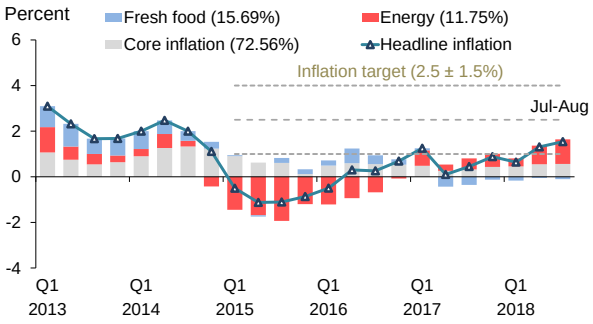
Nevertheless, the labor market still faced structural changes which affected wage adjustments, such as increased adoption of automation in place of human labor in the production process and labor mobility across manufacturing sectors which allowed labor shortages in one sector to be alleviated by labor from other sectors. Low inflation rates also allowed employers to avoid large wage increases. As a result, the average income of Thai workers remained at a low level relative to the past and to other countries in the region. Elevated household debt would also cause purchasing power to rise gradually in the period ahead, particularly in the case of households whose indebtedness rose faster than their income and thus, would be more likely to default on debt repayment.

Headline inflation increased from the previous quarter, mainly driven by energy prices. Core inflation remained stable with gradual increases in prices of goods and services.

Headline inflation averaged at 1.54 percent over the first two months of the third quarter of 2018, up from 1.31 percent in the previous quarter (Chart 2.2). The rise was mainly due to continued increases in energy prices stemming from higher domestic retail oil prices and LPG prices, in line with the direction of global oil prices. Meanwhile, fresh food prices declined from the previous quarter as prices of vegetables, fruits, and meats fell on account of high supplies in the markets.

Chart 2.2 Headline inflation increased from the previous quarter due mainly to energy prices.

Headline inflation and inflation target



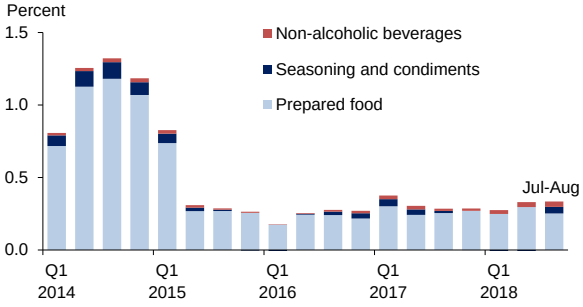
Note: ( ) denotes share in inflation baskets.
Source: Ministry of Commerce, calculations by Bank of Thailand

Core inflation averaged at 0.77 percent over the first two months of the third quarter of 2018, close to the previous quarter. Core inflation in the food category rose slightly (Chart 2.3) on the back of food seasoning and condiments and non-alcoholic beverages prices. Processed food price inflation declined slightly on account of lower fresh food costs and LPG price control by the government. Core inflation in non-food components dropped slightly (Chart 2.4) from slower increases in prices of vehicles and automobile parts after an earlier acceleration. In other categories, prices rose gradually as household purchasing power gradually improved. Moreover, structural factors including production technology development, rising trends of e-commerce, and globalization led to lower costs of production and intensified price competitions. As a result, these factors contributed to more persistent core inflation in non-food categories than in the past although the Thai economy expanded at its full potential.

Short-term (one-year ahead) inflation expectations according to the survey of businesses in August 2018 stood at 2.0 percent, largely unchanged from the previous quarter. Inflation expectations of professional forecasters slightly declined from the previous quarter to 1.4 percent. Long-term (five-year ahead) inflation expectations according to the survey of professional forecasters in April 2018 stood at 2.1 percent, up from 1.8 percent in the previous survey in October 2017.

Chart 2.3 Food-in-core inflation (28% of core inflation) increased slightly on the back of seasoning and condiments and non-alcoholic beverages prices.

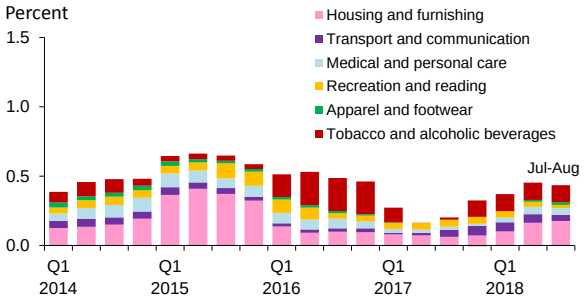
Contribution* to food-in-core inflation



Note: *Contribution to core inflation shows decomposition of core inflation changes according to weights of each good in core CPI basket.
Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by the Bank of Thailand

Chart 2.4 Non-food in core inflation (72% of core inflation) decreased slightly due mainly to a slowdown in vehicles and automobile parts price growth.

Contribution* to non-food in core inflation



Note: *Contribution to core inflation shows decomposition of core inflation changes according to weights of each good in core CPI basket.
Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by the Bank of Thailand

Table 2.1 Inflation

Annual percentage change	2016	2017				2018		
	Q4	Q1	Q2	Q3	Q4	Q1	Q2	Jul-Aug
Headline Consumer Price Index (Headline CPI)	0.69	1.25	0.10	0.45	0.88	0.64	1.31	1.54
Core Consumer Price Index (Core CPI)	0.73	0.66	0.47	0.49	0.61	0.61	0.76	0.77
Raw food	1.54	0.61	-2.99	-2.25	-0.80	-1.04	-0.35	-1.30
Energy	-1.06	6.69	2.67	4.86	5.24	3.01	7.30	9.62

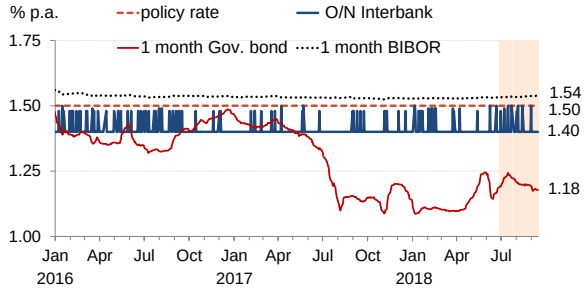
Source: Bureau of Trade and Economic Indices, Ministry of Commerce

Short-term money market rates remained close to the policy rate, except short-term government bond yields which continued to remain below the policy rate. Overall, government bond yields rose, especially medium-term bond yields, driven by better-than-expected recent economic outturns and increased amount of new bond issuance. Long-term government bond yields rose only slightly due to foreign investors' demand for certain government bonds for portfolio adjustments following changes in the benchmark bonds in the global bond index[2/].

Most **short-term money market rates** remained close to the policy rate in the third quarter of 2018. **Short-term (one-month) government bond yields** at the end of the third quarter rose from the previous quarter but remained below the policy rate (Chart 2.5). At the beginning of the quarter, short-term government bond yields were on the rise, partly on account of lower demand for treasury bills and higher supply of bonds issued by the Bank of Thailand. However, from the end of July, short-term government bond yields dropped slightly on account of higher demand as a large amount of Bank of Thailand bonds matured and investors shortened the duration of their bond holdings to reduce interest rate risk. **Yields on under ten-year government bonds** (Chart 2.6) fluctuated somewhat in the beginning of the third quarter on concerns over US trade protectionism, but started to rise until the end of the quarter on account of domestic factors including better-than-expected recent Thai economic outturns as well as increased issuance of government bonds in September. Yields on five-year government bonds rose higher than other durations, partly on account of investors selling their bonds ahead of the auction of five-year government bonds which were newly issued in addition to the Public Debt Management Office's plan announced at the beginning of the year. **Yields on over ten-year government bonds** rose only slightly on account of the issuance of new bonds for use as reference rate and J.P. Morgan's changes in Thai government bond series included in the global bond index. These factors contributed to investors' continued demand for long-term bonds in the third quarter.

Chart 2.5 Short-term interest rates remained close to the policy rate, except short-term government bond yields which moved at the low level.

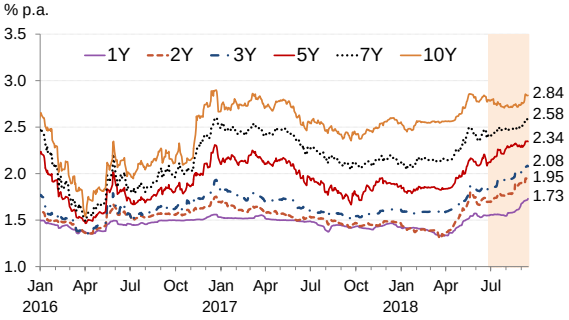
Short-term rates in financial markets



Sources: Bank of Thailand and Thai Bond Market Association (Thai BMA) (data as of 18 September 2018)

Chart 2.6 Short- and medium-term government bond yields increased following strong economic outturns and increased amount of newly issued bonds. However, long-term government bond yields increased only slightly due mainly to high demand by foreign investors for portfolio adjustment according to global bond index

Government bond yields



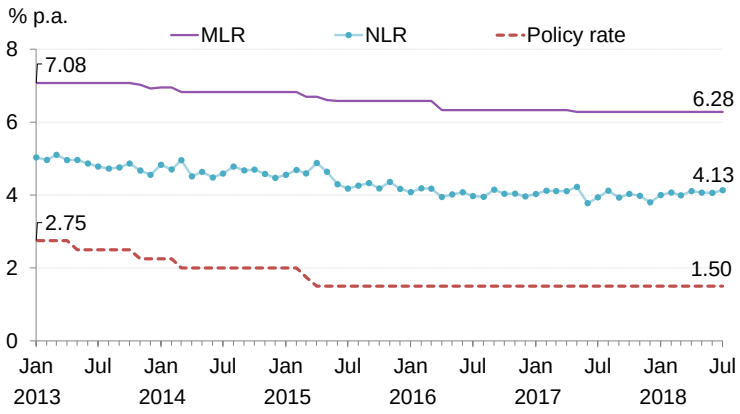
Source: Thai Bond Market Association (Thai BMA) (data as of 18 September 2018)

[2/] J.P. Morgan added five-year and thirty-year Thai government bonds in the GBI-EM Index (Global Bond Index-Emerging Markets)

Corporate bond yields remained low in line with stable credit spreads[3/]. **Cost of financing through commercial banks** as reflected in the new loan rate (NLR)[4/] also remained low, indicating financial conditions that remained accommodative (Chart 2.7).

Char 2.7 New Loan Rate (NLR) stabilized at low level

New Loan Rate



Source: Bank of Thailand (data as of July 2018)

Private credit extended to both businesses and households accelerated in line with continued economic expansion.

Private credit[5/] continued to accelerate in the second quarter and in July 2018 in line with the expansion of domestic demand. Private credit expanded at 5.9 percent in July 2018 from the same period last year (Chart 2.8), up from a 4.7 percent growth recorded at the end of the first quarter of 2018. Business credit growth accelerated, especially loans extended to SMEs, which had relatively large credit lines and were given to more diversified businesses such as real estate, construction, and public utilities. Meanwhile, loans extended to large corporates continued to improve in several businesses such as real estate, construction, commerce, and services. Household credit continued to improve across all loan purposes, particularly auto leasing loans.

3/ A credit spread is the difference between corporate and government bond yields with the same tenure, reflecting an assessment on corporate bond issuers' default risks.

4/ NLR is calculated based on a weighted average of interest rates for new loan contracts extended by 14 Thai commercial banks (excluding consumer loans and loans to financial intermediaries). The data covers loans of value of 20 million baht or higher for all purposes and terms and includes both secured and non-secured loans. Moreover, interest rates used in the calculation refer to the mid-rate between the lowest and the highest rates in each loan contract.

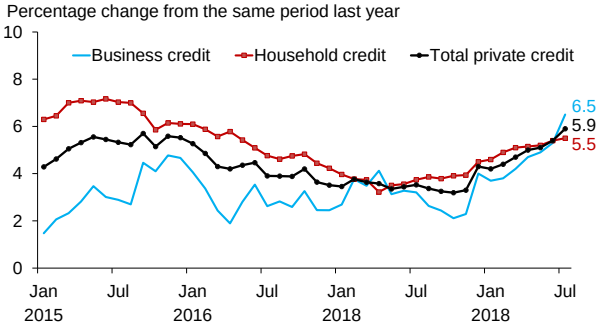
5/ Outstanding credit of other depositary corporations (ODCs), namely commercial banks, specialized financial institutions, finance companies, savings cooperatives, and market mutual funds.

The **net issuance of corporate bonds** continued to increase in the second quarter of 2018 from the previous quarter, mainly driven by funding of companies in the agribusiness, food and beverages, and energy sectors. The net issuance of corporate bonds declined somewhat in July 2018 from the end of the second quarter owing to redemptions by some issuers. However, the net corporate bond outstanding expanded 10.6 percent from the same period last year (Chart 2.9). **Funding through the equity market** continued to increase in the second quarter of 2018 and in July 2018, especially in businesses related to media and publishing, petrochemical and chemical, IT and communication, and commerce sectors.

Chart 2.8 Private credit accelerated from both business credit and household credit

Growth of private credit

Percentage change from the same period last year



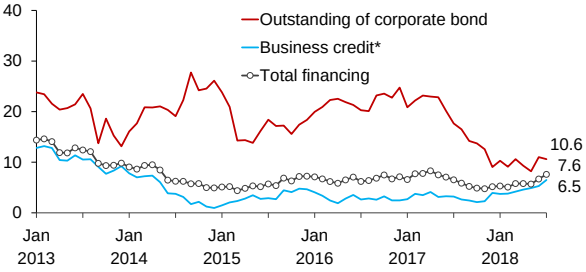
Note: Private credit includes credit to other depository corporations (ODCs) namely commercial banks, specialized financial institutions, finance companies, saving cooperatives, and money market mutual funds

Source: Bank of Thailand

Chart 2.9 Overall financing continued to expand

Growth of corporate bond outstanding and business credit

Percentage change from the same period last year



Note: *Business credit covers lending activities of Other Depository Corporation (ODCs)

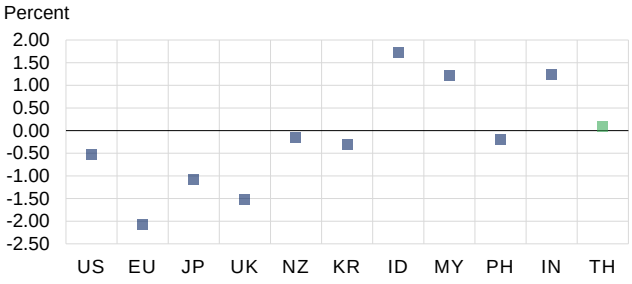
Sources: Thai Bond Market Association (Thai BMA) and Bank of Thailand

Going forward, financial conditions were expected to remain accommodative. Despite a slight increase from the previous *Monetary Policy Report*, the real policy interest rate remained accommodative overall and was moderate compared with other countries (Chart 2.10). Meanwhile, costs of financing through commercial banks, as reflected in the new loan rate (NLR) (Chart 2.7), would stabilize at a low level. According to the Credit Condition Survey[6]/, financial institutions were expected to maintain their credit standards for loans extended to large corporates, SMEs, and households in the third quarter of 2018 with exception of credit card loans, in which some financial institutions would be somewhat more vigilant.

Chart 2.10 Thailand's real policy rate* slightly increased but remained accommodative overall and was moderate compared with other countries

Real policy rates*

Percent



Note: *Calculated from policy rate subtracted by one-year-ahead inflation expectation according to a survey by Consensus Economics (as of 10 September 2018)

Sources: Bloomberg, Consensus Economics, calculation by Bank of Thailand

[6]/ Survey of credit conditions for the second quarter of 2018 and outlook for the third quarter of 2018.

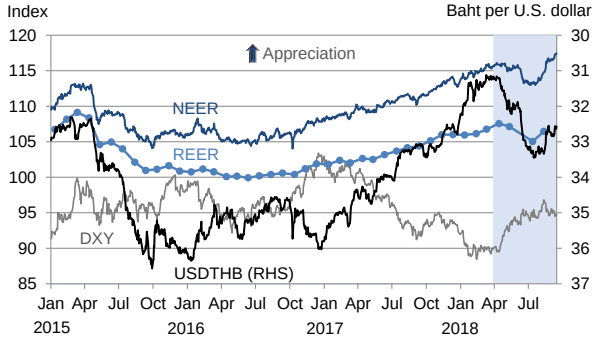
The baht appreciated against the U.S. dollar and the nominal effective exchange rate also appreciated, mainly supported by domestic factors.

In the third quarter of 2018, the **baht appreciated against the U.S. dollar** relative to the end of the previous quarter (Chart 2.11). In the first half of the third quarter, the baht weakened slightly against the U.S. dollar in line with movements of regional currencies. This was attributed to the U.S. dollar's continued strengthening following investors' concerns over US trade protectionism and retaliatory measures from its trading partners which could further intensify. In addition, investors continuously reduced their risky asset holdings in emerging markets (EMs) as a consequence of fragile economic fundamentals problems in some emerging market economies such as Turkey and Argentina. However, since the middle of August 2018, the baht appreciated against the U.S. dollar, mainly attributed to domestic factors. Such factors included Thailand's economic outturns which were better than expected, prompting markets to anticipate sooner-than-expected Thai policy rate hike. Moreover, there was greater clarity on the timeline of the general election in Thailand. Meanwhile, investors' easing concerns over US trade protectionism and measures implemented by emerging market economies to cope with capital outflows and plummeting currencies prompted investors to return their investments to emerging market economies which had strong external balance such as Thailand, Korea and Taiwan. Consequently on 18 September 2018, the baht closed at 32.57 to the US dollar, up 1.7 percent from the end of the previous quarter.

The **nominal effective exchange rate (NEER)** index stood at 117.44 on September 18, 2018, a 3.6 percent appreciation from the end of the previous quarter. The movement was in line with the baht appreciation against currencies of most trading partners, particularly the Chinese yuan, regional currencies, and currencies of emerging market economies which depreciated amid foreign investors' selloffs of emerging market assets (Chart 2.12). As of the end of August 2018, the **real effective exchange rate (REER)** rose 2.5 percent from the end of the previous quarter. In the period ahead, exchange rates would likely remain volatile due to uncertainties over monetary-fiscal policies and trade policies of major economies, geopolitical risks, and the outlook for oil prices.

Chart 2.11 The baht depreciated against the U.S. dollar as the U.S. dollar strengthened

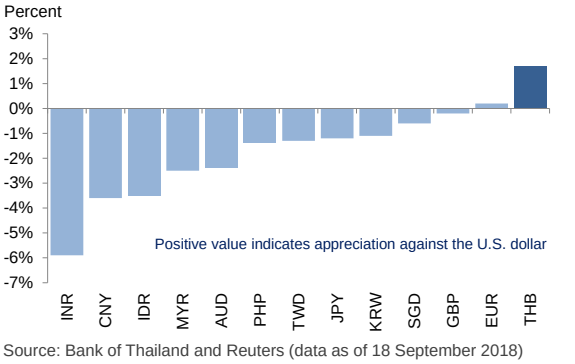
USDTHB, NEER, DXY



Source: Bank of Thailand and Reuters (data as of 18 September 2018)

Chart 2.12 Major currencies and most regional currencies depreciated against the U.S. dollar.

Currency changes against the U.S. dollar (18 Sep 18 compared to 30 Jun 18)



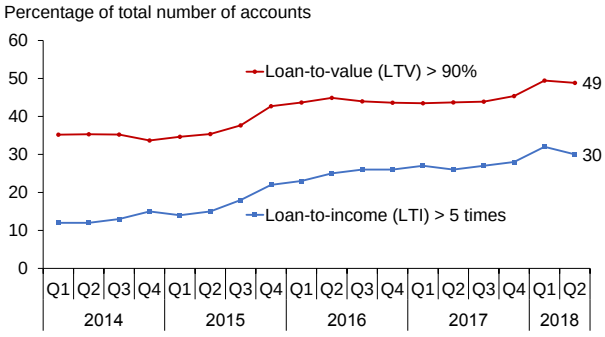
Source: Bank of Thailand and Reuters (data as of 18 September 2018)

Financial stability remained sound overall. However, there remained pockets of risks that warranted monitoring including more fragility in the real estate sector, elevated household debt with worsening debt serviceability of certain households and small enterprises, continued search-for-yield, and volatilities in global financial markets which could affect financing costs of the private sector.

Thailand's financial stability remained sound overall, especially external stability as reflected in the country's high level of international reserves and sustained current account surplus, while the external debt to GDP ratio remained low[7]/. These factors combined to cushion the Thai economy against recent volatilities in the global financial markets. Financial institutions maintained strong financial positions, as reflected in high levels of capital buffers of commercial banks to cushion against risks stemming from deterioration of credit quality. Nevertheless, there remained pockets of risks that warranted monitoring going forward. Such risks were as follows.

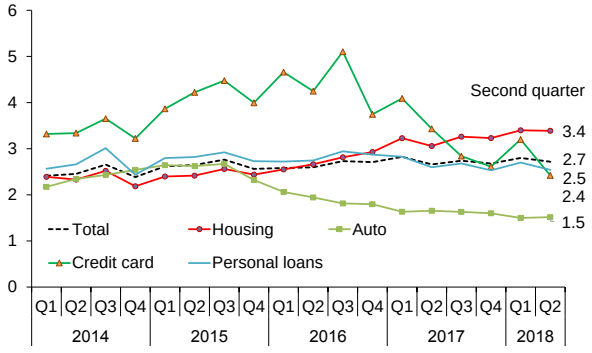
(1) Signs of increasing in vulnerabilities in the real estate sector as financial institutions' competition in mortgage loan extension with a willingness to bear higher risks had resulted in loosening credit standards. This was reflected in a rising share of commercial banks' new mortgage loan accounts with the loan-to-value (LTV) ratio exceeding 90 percent as well as a rising share of accounts with high loan-to-income (LTI) ratios (Chart 2.13). The ratio of loan extension for a second or more home purchases also increased with loosening credit standards. Moreover, the quality of mortgage loans continued to deteriorate as indicated by a continuing uptrend of the non-performing loan (NPL) ratio in mortgage loans (Chart 2.14). Thus, such vulnerabilities warranted close monitoring. However, the situation of accumulated excess supply abated somewhat as reflected in a decline in both the number of condominium inventory and the time taken for all units to be sold at the end of the first half of 2018 compared with 2017. This was partly because developers delayed new project launches. The oversupply in some areas still, however, warranted monitoring, especially of the projects along the Purple Line, since developers were expected to launch new projects rapidly in the first half of 2018 before the enactment of the new Nonthaburi's city planning law.

Chart 2.13 Financial institutions competed in extending mortgage loans and were willing to bear higher risks.
Ratio of number of newly approved housing loans account, classified by loan-to-value (LTV) and loan-to-income (LTI)



Source: Bank of Thailand

Chart 2.14 The non-performing loan (NPL) ratio in mortgage loans was expected to trend up.
The NPL ratio of household credit, classified by purposes
%NPL of each type of loan



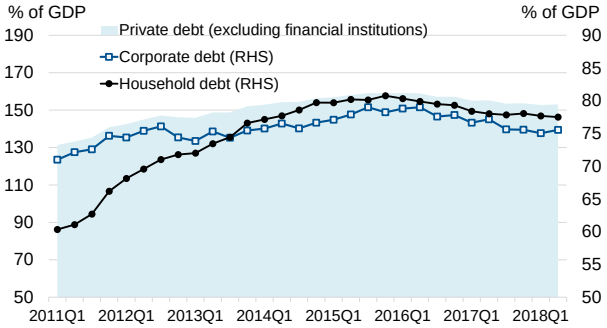
Source: Bank of Thailand

[7]/ 33.4 percent as of the latest data in the second quarter of 2018

(2) Household debt remained high, while debt serviceability of some households and small enterprises deteriorated. Although the downtrend in the ratio of private debt to GDP (Chart 2.15) indicated a decline in risks from new debt accumulation, household debt which remained high and had yet to show clear signs of deleveraging implied households' limited capability of resilience against economic volatilities. Besides, persistently low interest rates induced households to incur new debt, and that could weigh on consumption and debt serviceability (Box: *Implications of household debt on the Thai economy and financial system stability*). Overall credit quality remained largely unchanged as indicated by the NPL ratio of commercial banks in the second quarter of 2018 which was stable at 2.93 percent. However, financial positions of some small enterprises remained vulnerable as reflected in the sustained negative operating profit margins (OPM) and the interest coverage ratio (ICR) of some small businesses [8/]. These were consistent with the NPL ratio of commercial banks' SME loans which remained high at 4.5 percent[9]/.

Chart 2.15 The ratio of private debt to GDP trended downward since the second half of 2016

The ratio of private debt to GDP

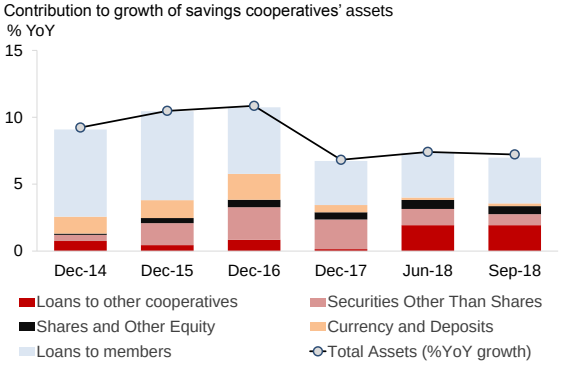


Source: Bank of Thailand

(3) The continued search-for-yield behavior under the environment of persistently low interest rates could lead to underpricing of risks. Although systemic risks remained limited, there remained issues that warranted monitoring including, first, a search-for-higher-yield behavior of saving cooperatives that provided high rates of return to their members. As a consequence, saving cooperatives' assets and deposits continued to expand at a high rate, putting pressure on them to search for higher yield (Chart 2.16). Nevertheless, the growth rate slowed down somewhat after regulatory authorities collaborated to enhance supervision standards. In the meantime, credit risk and liquidity risk warranted monitoring due to saving cooperatives' borrowing among each other and substantial investment in debt instruments. In addition, some large saving cooperative increased borrowing from commercial banks and lent to other saving cooperatives, resulting in overall increased systemic risks in the saving cooperatives sector and the Thai financial system. Second, businesses and households turned to alternative investment channels and accepted higher risks. For instance, under the low interest rate environment, large corporates increasingly raised funds for non-core business investments and households invested more in the real estate for speculative or rental purposes. Third, although offshore investments

Chart 2.16 Saving cooperatives are more inclined to search for yield behavior, especially in debt securities and loans to other cooperatives

Contribution to growth of savings cooperatives' assets



Note: Savings cooperatives were subjected to tighter regulation by government since H2/2017

Source: Cooperative Auditing Department, calculations by Bank of Thailand

[8/] As reflected by the ICR and OPM of 25th-percentile small-sized listed companies which have been in the negative territory since 2013.

[9/] The average ratio of NPL for SMEs during 2013-present was 3.8 percent.

through foreign investment funds (FIF) had contracted in the past few months on concerns over volatilities in global financial markets, concentration risks remained as investments by FIFs were concentrated in five major countries.

(4) Volatilities related to capital flows and global financial markets could trigger yield snapbacks and affect debt rollovers of companies that raised funds through short-term bond issuance. Although over 80 percent of corporate bonds issued had A or higher credit rating and 85 percent of the issues were long-term baht-denominated bonds, the upward trends in interest rates in global financial markets could trigger yield snapbacks and affect rollovers in the period ahead, particularly in the case of companies that issued bonds with credit rating below A and remaining maturity less than one year. Such bonds stood at over 200 billion baht outstanding in the second quarter of 2018.

2.2 Outlook for the Thai Economy

Under the Committee's assessment, Thailand's economic growth was projected to gain further traction. The Thai economy was expected to grow at 4.4 percent and 4.2 percent in 2018 and 2019, respectively, in line with the assessment in the previous *Monetary Policy Report*. Key growth drivers would be from (1) private spending including consumption and investment, (2) merchandise exports, consistent with growth in global trade volume and trading partners' economies, (3) tourism, consistent with the outlook of foreign tourist arrivals, and (4) public spending, despite somewhat lower-than-expected growth. Inflation was expected to trend up in line with the previous assessment, while fresh food prices would be lower than expected.

Table 2.2 Forecast summary

Percent	2017*	2018	2019
GDP growth	3.9	4.4 (4.4)	4.2 (4.2)
Headline inflation	0.7	1.1 (1.1)	1.1 (1.2)
Core inflation	0.6	0.7 (0.7)	0.8 (0.9)

Note: * Outturn
( ) Monetary Policy Report June 2018
Sources: NESDB, Ministry of Commerce, Bank of Thailand's estimates

Summary of the key forecast assumptions

- Trading partner economies would expand at lower rates than expected throughout the forecast horizon due to intensifying trade protectionism. The growth projection took into account some impacts of the U.S.-China trade protectionism measures whereby the U.S. announced to impose 200 billion U.S. dollars tariffs on import goods from China. Moreover, growth rates in Asian economies were expected to be lower as some countries tightened monetary policy to preserve price and external stability. Lower growth rates would also be expected for economies in the euro area mainly due to a private consumption slowdown.

- The federal funds rate projection remained unchanged from the previous one. The Fed was expected to raise the policy rate in total 4 times in 2018 and 3 times in 2019, while gradually commencing its balance sheet reduction according to announced plan.

- Asian currencies (excluding the Chinese yuan), throughout the forecast horizon, were revised to be weaker than the previous assessment after the outturns of Asian currencies during the third quarter were weaker than expected as a result of currency crisis in some emerging market economies. In 2019, Asian currencies were expected to gradually appreciate from the end of 2018 due to overall robust economic conditions in Asia.

- The Dubai crude oil price was revised up throughout the forecast horizon after oil supply was lower than expected as supply from Iran fell following US sanctions. Moreover, supply from Venezuela temporarily dropped after the tanker collision at the oil port. In the period ahead, the Dubai crude oil price was projected to gradually decrease to an equilibrium level in line with global economic fundamentals.

• Farm income was slightly revised up throughout the forecast horizon on the back of higher-than-expected output in many products including rice, sugarcane, cassava, and rubber.

• Public spending at current prices was revised down for both consumption and investment expenditure. Government consumption was revised down due to compensation of civil servants given policy to replace vacant job positions with contract workers. Public investment was revised down for both the central government and state-owned enterprises. The central government's investment projection would decrease due to lower-than-expected efficiency in budget disbursement following construction problems and larger-than-expected impacts of the Public Procurement and Supplies Management Act, B.E. 2560. For state-owned enterprises, investment projects on land transportation could be delayed owing to construction issues and funding reviews.

Table: Summary of forecast assumptions

Annual percentage change	2017*	2018	2019
Dubai crude oil price (U.S. dollar per barrel)	53.1	70.3 (69.2)	69.8 (68.3)
Farm income (% YoY)	1.5	2.9 (2.8)	2.3 (2.1)
Government consumption at current price (billion baht)[1/]	2,532	2,656 (2,669)	2,786 (2,819)
Public investment at current price (billion baht)[1/]	926	992 (1,028)	1,082 (1,115)
Fed funds rate (% at year end)	1.38	2.38 (2.38)	3.13 (3.13)
Trading partners' GDP growth (% YoY)[2/]	3.9	3.8 (3.8)	3.5 (3.6)
Regional currencies (excl. China) vis-à-vis the U.S. dollar (index)[3/]	155.7	153.2 (149.9)	153.8 (149.1)

Notes: [1/] Assumption includes spending on infrastructure investment plans
[2/] Weighted by each trading partner's share in Thailand's total exports
[3/] Increasing index represents depreciation, decreasing index represents appreciation
* Outturns
( ) Monetary Policy Report June 2018

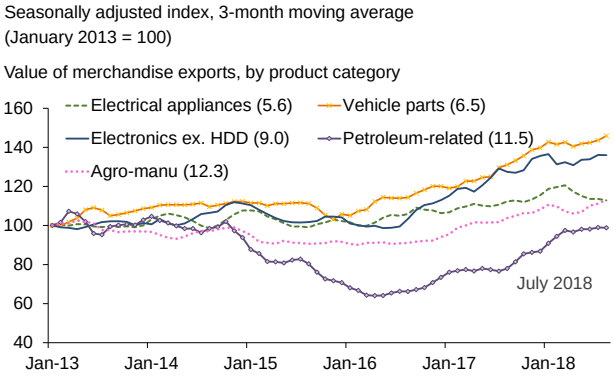
The value of merchandise exports in 2018 was projected to be in line with the previous assessment, but the projection for 2019 was revised down given some impacts from the U.S.-China trade protectionism measures.

Thai exports were expected to record robust growth in both 2018 and 2019, partly due to a special factor involving the relocation of production base to Thailand for some export-oriented industries such as hard disk drives. **The value of merchandise exports in 2018 was projected to expand at 9.0 percent, unchanged from the assessment in the previous *Monetary Policy Report*.** While export volume was revised up with the outturns in the second quarter, export prices in U.S. dollar term would display a slower growth following the baht's depreciating trend which was in line with the movement of regional currencies. As a result, the growth rate in value of exports was projected to be similar to the previous assessment. Recent Thai exports exhibited impressive growth across most product categories and most export destinations, except for declining exports of electrical appliances to the U.S. following the impact of US trade protectionism measures on imports of solar cells and large washing machines (Chart 2.17 and 2.18). **For 2019, growth in the value of exports was expected to slow down to 4.3 percent from the previous estimate of 5.0 percent** due to both price and volume factors. The revision was due to some expected impacts of US trade

protectionism measures and its trading partners' intensifying retaliatory measures on global trade volume and Thailand's trading partner economies.

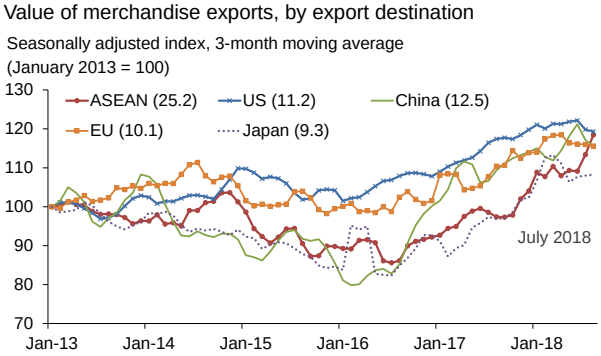
The Committee assessed that trade protectionism measures between the U.S. and major economies could intensify and rapidly develop. This would put pressures on international trade and investment, which would in turn affect Thai exports in both direct and indirect ways. Thus, the Committee would monitor closely developments of trade policies and negotiations, their effects on supply chains, and trade diversion which could have impacts on Thai businesses.

Chart 2.17 Merchandise exports continued to expand across various product categories except for declining exports of electrical appliances to the U.S. following the impacts of US trade protectionism measures on imports of solar cells and large washing machines.



Note: Number in () denotes share to total exports in 2017
Source: Customs Department, calculation by Bank of Thailand.

Chart 2.18 Merchandise exports expanded across almost all export destinations



Note: Numbers in ( ) represent share in total exports in 2017
Source: Customs Department, Calculations by Bank of Thailand

Exports of services in 2018 were revised down due to a temporary factor but next year would see higher growth than previously assessed due to a robust tourism outlook.

In 2018, overall growth rate in exports of services was lower than the previous assessment as non-tourism services slowed down, although the projected number of foreign tourists in 2018 was at 38.3 million, unchanged from the assessment in the previous *Monetary Policy Report*. The higher-than-expected number of tourists in the second quarter reflected that Thailand remained a major destination for foreign tourists. The figure was in line with greater sentiment among travelers, especially for free and independent travelers (FIT) who had high purchasing power. This would help offset a temporary drop in the number of Chinese tourists during the second half of 2018 following the impact of Phuket tour boat sinking incident in early July. The impact of the Phuket incident was expected to affect only coastal tourism destinations and to be short-lived as the number of Chinese tourists began to see some recovery in many areas. The improvement was partly thanks to intensive marketing campaigns in China, elevated safety standards by Tourism Authority of Thailand, as well as the private sector's sales promotion targeted at Chinese tourists. **In 2019, exports of services were projected to exceed the previous estimate on account of improvement in tourism's outlook. The projected number of foreign tourists in 2019 was revised up to 40.6 million from the previous assessment of 40.0 million.** The upward revision was partly due to postponed travel plans by some Chinese tourists from the second half of 2018 and improvement in other supporting factors including tourist confidence and the openings of new airline routes from ASEAN to Thailand. Another factor was capacity management by

airlines and airports, such as additional flights during off-peak times and greater utilization of regional airports, which would enhance tourism carrying capacity.

The current account was projected to register a 35.4 billion U.S. dollar surplus in 2018, down from the estimated surplus of 40.0 billion U.S. dollars in the previous *Monetary Policy Report*. The revision was due to the downward revision of the value of service exports and the upward revision of the value of import goods and services during the third quarter. In particular, gold imports increased as gold prices dropped sharply. **The projection for the current account in 2019 was 36.3 billion U.S. dollar surplus, close to the previous estimated surplus of 36.0 billion U.S. dollars.**

Private consumption growth was projected to be higher than the previous assessment due to improvements in household income.

Private consumption was expected to gain further traction. Supporting factors were from a steady growth in non-farm income among middle- to high-income households and income growth among low-income households following improvements in employment in most sectors. Farm income also expanded on the back of higher agricultural production and support from government policies such as the first and second phases of the social welfare card scheme, the community enterprise development project, and the agricultural reform project. However, elevated household debt would cause households to allocate part of their income for debt repayment. Moreover, structural changes in the labor market limited wage increases. These changes included, first, greater adoption of automation in place of human labor in the production process. Second, sectoral labor mobility allowed labor shortage in one sector to be alleviated by labor from another sector. Third, low inflation would in turn lessen upward pressures on wages. As a result, average wages of Thai workers remained low and purchasing power would recover gradually in the period ahead.

Public spending would help drive the economy despite some slowdown when compared with the previous assessment.

Public spending was expected to be lower than the previous assessment. Government consumption expenditure, particularly compensation of civil servants with regard to salary and medical expenditure, was projected to decrease given the policy to replace vacant job positions with contract workers. Public investment decreased for both the central government and state-owned enterprises. For the central government, investment was revised down due to lower efficiency in budget disbursement by some government units, following construction problems that included limited construction capacity, land reclamation issues, and larger-than-expected impacts of the Public Procurement and Supplies Management Act, B.E. 2560. For state-owned enterprises, some investment projects faced operational difficulties. For instance, the SRT red line[10/] and the suburban railway projects (Bang Sue-Rangsit) would see some delays due to project revisions. The Purple Line project (Tao Poon-Rat Burana) was subject to delays following funding reviews. The Rama III-Daokanong expressway project also faced some bidding process issues.

[10/] The SRT light red line (Bang Sue-Phaya Thai-Makkasan-Hua Mak) and the SRT dark red line (Bang Sue-Hua Lam Pong)

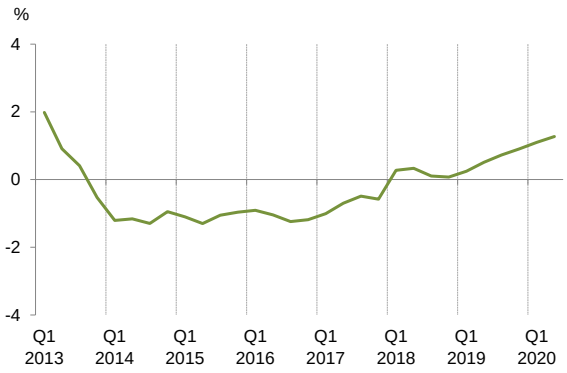
Private investment would see more robust growth in the period ahead.

Private investment in 2018 was projected to grow in line with the previous assessment, while investment growth in 2019 was expected to be slightly higher than the previous estimate. The slight upward revision was due to better-than-expected private consumption, together with capital outlays following production relocation to Thailand of some export-oriented industries during late 2018 and 2019. Robust investment outlook was reflected in (1) higher capacity utilization in various industries such as automobiles, electronics, and chemical products, (2) greater clarity on investment plans of large companies, (3) higher demand for corporate credits, and (4) improved business sentiment. Positive private investment outlook would also be supported by the continuation of public infrastructure investment projects, the Eastern Economic Corridor (EEC), and public-private partnership (PPP). In addition, joint investment by various groups in the private sector in the bidding for the mass rapid transit projects and the high-speed trains connecting three airports[11]/ would boost business confidence, foster investment climate, and attract greater foreign investment. Nonetheless, project developments and feasibility would warrant monitoring going forward.

Inflation was expected to rise in line with the previous assessment.

In the period ahead, inflation was expected to rise in line with the assessment in the previous *Monetary Policy Report*. Price increases in fresh food items were projected to be lower than expected mainly because the supplies of meat, vegetables, and fruits were expected to increase more than previously assessed thanks to favorable weather conditions and the government's water management that gave priority to agricultural purposes. Meanwhile, energy price inflation rose more than expected in tandem with global crude oil prices. Demand-pull inflationary pressures lowered slightly due to the weakened relationship between the output gap (Chart 2.19) and inflation. This was partly due to ongoing sales promotion and structural factors that still weighed on inflation, including improvement in production technology that resulted in lower costs of goods and services as well as greater competition that constrained businesses from raising prices. **The Committee therefore projected headline inflation to average at 1.1 percent in both 2018 and 2019 and core inflation to average at 0.7 and 0.8 percent in 2018 and 2019, respectively.**

Chart 2.19 Output Gap



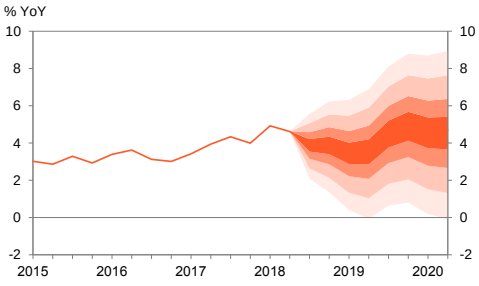
Risks to the growth projection were expected to tilt downward but with a smaller degree than the previous assessment. Meanwhile, risks to the inflation forecast were expected to tilt downward in line with risks to the growth projections.

Under the Committee's assessment, risks to the growth projection were expected to tilt downward but with a smaller degree than the assessment in the previous *Monetary Policy Report*, as reflected in the fan chart with smaller downside

[11]/ Construction expected to commence during the latter half of 2019

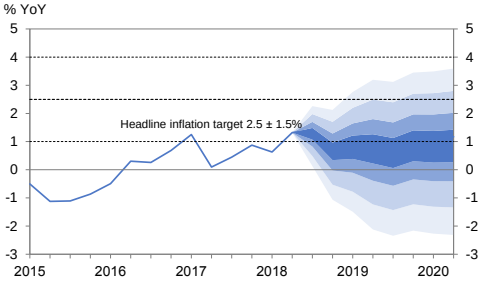
skewness (Chart 2.20), due to a potential upside from government's measures aimed at stimulating private spending which could be additionally announced. On the other hand, **possibilities that the Thai economy would underperform the baseline projection remained**. The first possibility involved US trade protectionism measures and additional retaliatory measures from major economies despite the baseline projection already taking into account some impact of the 200 billion dollar worth of tariff that the U.S. imposed on China's exports. Moreover, intensifying competition resulted from trade diversion could have greater-than-expected impacts on Thailand's exports and private investment. Second, Thailand's trading partner economies could expand at lower rates than expected in case of intensifying global geopolitical tensions and economic problems among emerging market economies. Third, gradual improvements in domestic purchasing power could result in a lower-than-expected private consumption growth. Fourth, the enforcement of the Public Procurement and Supplies Management Act could delay budget disbursements of some government agencies. Moreover, risks remained as state-owned transportation enterprises might experience delays in their investment plans due to reviews of investment project approvals. Nevertheless, there was **possibility that the Thai economy would outperform the baseline projection.** First, growth of Thailand's trading partner economies could be higher than expected on account of continued improvements in the U.S. economy with support from tax reforms. The Chinese economic slowdown could also be less severe than expected if the Chinese government were to announce additional stimulus measures, which would eventually lead to better-than-expected Asian exports. Second, domestic spending could be higher than expected due to infrastructure investment projects, public–private partnership (PPP), and government measures aimed at stimulating private spending that could be additionally announced in the period ahead. And third, the number of Chinese tourists could be larger than expected following the recovery after the Phuket tour boat sinking incident. Meanwhile, **risks to the forecasts of headline and core inflation were expected to tilt downward** in line with risks to the growth projections and lower-than-expected fresh food prices amid volatile weather conditions and production which could be better than expected (Chart 2.21 and 2.22).

Chart 2.20 Growth forecast



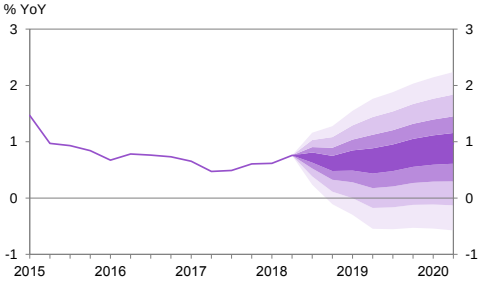
Note: Fan chart covers 90% of the probability distribution

Chart 2.21 Headline inflation forecast



Note: Fan chart covers 90% of the probability distribution

Chart 2.22 Core inflation forecast



Note: Fan chart covers 90% of the probability distribution

Table 2.3 Forecasts of GDP and components

Annual percentage change	2017*	2018	2019
GDP growth	3.9	4.4 (4.4)	4.2 (4.2)
Domestic demand	2.1	3.9 (3.9)	3.8 (3.8)
Private consumption	3.2	4.2 (3.7)	3.7 (3.6)
Private investment	1.7	3.7 (3.7)	4.5 (4.4)
Government consumption	0.5	2.3 (2.7)	2.2 (2.9)
Public investment	-1.2	6.1 (8.9)	7.7 (6.5)
Exports of goods and services	5.5	5.5 (5.5)	4.1 (3.8)
imports of goods and services	6.8	7.5 (6.3)	3.3 (3.8)
Current account (billion, U.S. dollars)	51.1	35.4 (40.0)	36.3 (36.0)
Value of merchandise exports	9.8	9.0 (9.0)	4.3 (5.0)
Value of merchandise imports	13.2	16.9 (14.7)	5.6 (6.9)
Number of foreign tourists (million person)	35.4	38.3 (38.3)	40.6 (40.0)

Note: *Outturns

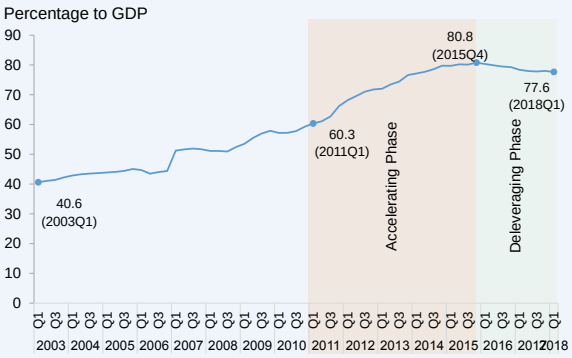
( ) Monetary Policy Report June 2018

Implications of household debt on the Thai economy and financial system stability

Household debt[12/] has received much attention from the public for the past decade as it accelerated significantly since 2011[13/]. This is reflected in the ratio of household debt to GDP[14/] that increased from 60.3 percent at the end of the first quarter of 2011 to 80.8 percent at the end of the fourth quarter of 2015. Although households have begun to slowly deleverage since the beginning of 2016, the ratio of household debt to GDP remains high (Chart 1) and is at the top of the rankings among the region (Chart 2). As a result, this has led to rising concerns that Thailand's elevated household debt might affect private consumption and derail economic growth in the long run. Moreover, it might have an impact on debt serviceability of households and also increase financial stability risks. This article thus focuses mainly on two issues including implications on the Thai economy and financial system stability.

Chart 1 The ratio of household debt to GDP ratio remains elevated.

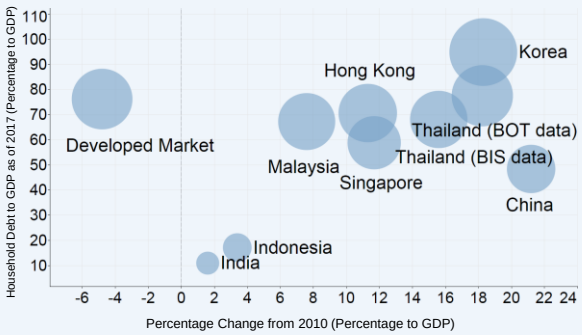
Household debt to GDP in Thailand



Source: Bank of Thailand

Chart 2 The ratio of household debt to GDP in Thailand at the top of the rankings among the region.

Household debt to GDP as of 2017 and percentage change from 2010



Note: Household debt data is derived from the Bank for International Settlements (BIS data) which is on a comparable basis between countries. Compared to BIS data, the data from BOT (BOT data) has wider coverage of creditors (includes other financial corporations, e.g., non-bank) which results in a higher ratio of household debt to GDP ratio.
Source: Bank for International Settlements (BIS) and Bank of Thailand

Implications on the Thai economy:

Household debt causes private consumption to expand at a lower rate than it should be

Households play an important role in driving and supporting economic growth by acting as "savers," passing on their savings through financial institutions to those who need funds, which are firms that borrows money to invest in the production of goods and services as well as households in need of funds. At the same time, households also play a role as "consumers" of goods and services that are produced by firms. In terms of value, private consumption accounts for about half of Thailand's GDP (Chart 3). However, such ratio has been declining

12/ Household debt in this article covers only debt extended from financial institutions to residents. Financial institutions, hereby, refer to (1) depository institutions such as commercial banks, specialized financial institutions accepting deposits, savings cooperatives, and (2) other financial institutions such as credit card companies, leasing companies, personal loan companies, insurance and life insurance companies, and pawnshop.

13/ There are several reasons such as a better access to borrowing sources in the system, innovation and diversified financial products, debt accumulation to fix damages after the flood disaster in 2011, debt accumulation following the first-time car buyer scheme, debt accumulation by farmers due to severe droughts.

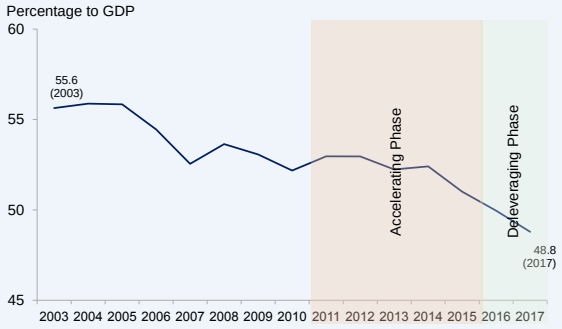
14/ Most studies use the ratio of debt to GDP as a reflection of a ratio of debt to income because GDP data are comparable across countries and released before household disposable income data.

especially since 2016. This is due to factors related to both the business cycle and structural changes[15/] such as labor migration out of the manufacturing sector to the services sector with relatively lower wages, a transition toward aging society that has prompted households to save more for retirement, and elevated household debt.

Debt creation allows households to increase spending in the short run and is an alternative way for consumption smoothing in case of economic fluctuations. On the contrary, debt creation increases debt burden which might decrease households' ability to spend in the future should such debt accumulation is not used for income-generating activities. Nevertheless, a recent study on impacts of household debt on Thailand's private consumption[16/] suggested that the **rise in household debt helped boost consumption in the short run but was found to be a factor holding down growth in the long run**, which is defined in this study as four years later.

Chart 3 Private consumption accounts for about half of the GDP.

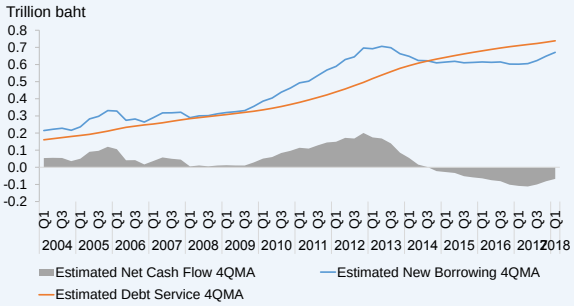
Thailand's private consumption to GDP



Source: Office of the National Economic and Social Development Board, calculated by Bank of Thailand

Chart 4 Household debt currently remains a factor causing private consumption to expand at a lower rate than it should be.

Net cash flow from debt creation to the economy



Note: Net Cash Flow is the difference between New borrowing and Debt service
Source: Bank of Thailand

Net cash flow generated from *debt creation*, assuming no changes in net cash flow from *income*, can be calculated from the difference between the value of new debt and the value of existing debt that is repaid. It is found that net cash flow generated from household debt creation was still negative in the first quarter of 2018 (Chart 4, grey area), meaning less money for households to spend. This is one of the reasons why **private consumption has expanded at a lower rate than it should be**. However, such effect has started to be lessen since the previous year as households increase debt accumulation. This is particularly observed in auto leasing, driven by demand for new cars as the impact from the first-car buyer scheme, where auto possessions needs to be maintained for five years, gradually dissipated. However, although debt creation helps boost household spending, if accelerated, it would have implications for financial stability in the household sector.

15/ Veenussaya and Chalawas (2018) "Investigating Thailand's private consumption: Why growth is not as high as in the past?" FAQ Issue 135.

16/ Suwanik S. and Peerawattanachart K. (2018) "Household debt in SEACEN Economies: Thailand" SEACEN Research Paper.

Implications on financial system stability

Households are subject to risks from heavy debt burden, deteriorating debt serviceability, and limited capacity to cope with shocks

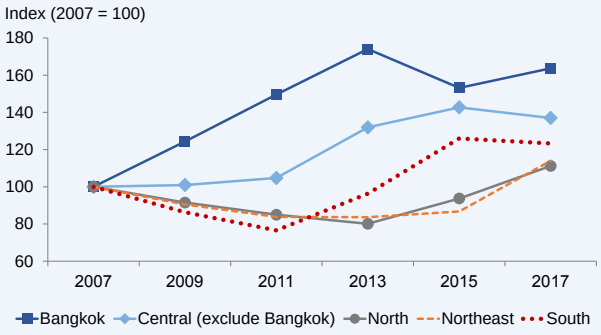
Debt creation is one of risk management tools, for instance, to cope with a severe drought. However, excessive debt creation can create vulnerabilities to households' balance sheets should households with debt burden have lower income due to various reasons such as economic slowdown and natural disasters. As a result, households might not be able to repay debt as scheduled, leading to higher non-performing loans (NPLs) of financial institutions and eventually an impact on overall financial system stability.

Analysis of the impact of household debt on Thailand's financial system stability consists of three aspects: (1) level and speed of leverage, (2) debt serviceability, and (3) ability to cope with income and interest rate shocks.

First, although the ratio of household debt to GDP slowly declined since early 2016 (Chart 1), the ratio remains high and there are signs of acceleration in household debt since the beginning of 2017[17/]. This is particularly observed in consumer loans for all spending categories especially auto leasing. Meanwhile, household income has not sufficiently increased in a broad-based manner, with only medium- and high-income households in the manufacturing and tourism sectors being the first group to see improvements in income. On the contrary, farm income recovers only gradually, with the majority of farmers having low income but substantial debt. In addition, **a decrease in the ratio of household debt to income is concentrated only in households in certain regions**, namely the central and southern regions. Meanwhile, other regions see the ratio of household debt to income increases (Chart 5).

Chart 5 Deleveraging is apparent only in some groups of households

Index of household debt to income (median), classified by region

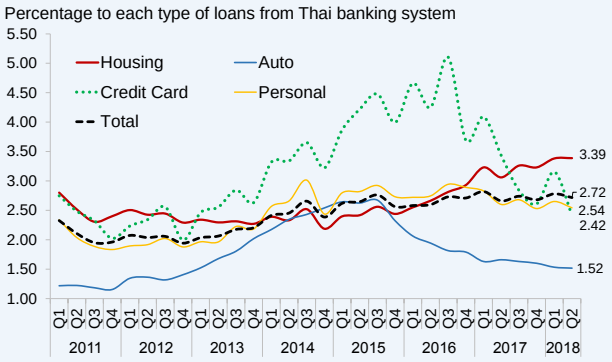


Source: Socio-Economic Survey, National Statistical Office, calculated by Bank of Thailand

Second, debt serviceability of some households continues to deteriorate and that warrants monitoring. The NPL ratio of consumer loans extended by commercial banks trended up since 2013 to 2.2 percent at the end of the second quarter in 2018, driven mainly by a continue rise in mortgage NPLs (Chart 6). Moreover, when classified by income and occupation, the debt service ratio (DSR) is found to increase for some groups, particularly the low-income and agricultural households (Chart 7).

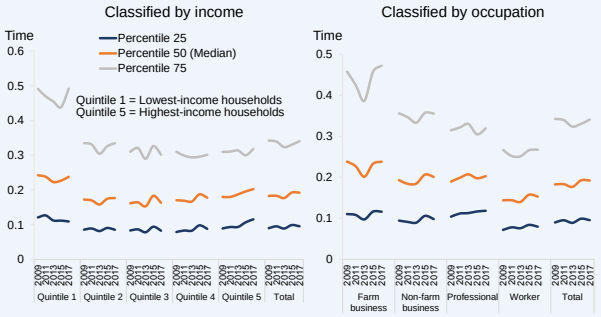
[17/] Considering a quarter-on-quarter growth of household debt after seasonal adjustments

Chart 6 Household debt serviceability continues to deteriorate.
Non-performing (NPL) of the banking system



Source: Bank of Thailand

Chart 7 Household's debt service ratio remains high.
Ratio of monthly debt service to monthly income (DSR)

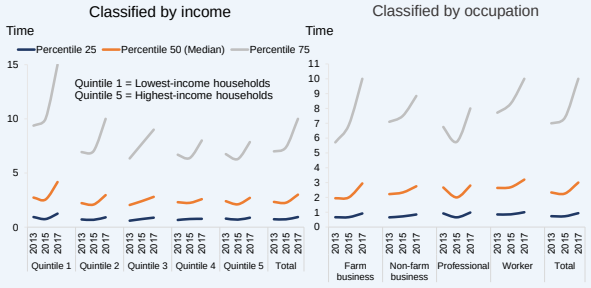


Note: Calculations include only indebted households.
[1/] Households are classified into five quintiles based on their income per capita, with the 1st quintile having lowest monthly income per capita and the 5th quintile having highest monthly income per capita.
[2/] "Professional" households include managers, academicians and professionals, technicians, etc.
[3/] "Worker" households include those working in agriculture, forestry, fishery, machine control, clerkship, services, craftsmanship, manufacturing operation, etc.
Source: Socio-Economic Survey, National Statistical Office, calculated by Bank of Thailand

Third, households have lower capacity to cope with economic and financial shocks. In particular, household financial positions become more susceptible to interest rate shocks as well as income shocks. Households have less cushion to cope with financial risks given high debt relative to savings, as reflected in the ratio of debt to financial assets (savings) that increases for all income and occupation groups (Chart 8). Moreover, **the household sector is found to be slightly more sensitive to interest rate shocks,** as reflected in the marginal increase in the ratio of debt with floating rates in comparison to 2014 (Chart 9).

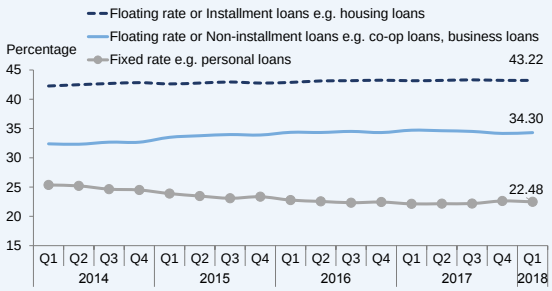
However, higher interest rates in the period ahead are expected to have limited impact on monthly debt repayment of households. This is because, first, installment loans, whereby borrowers pay a fixed monthly payment with a flexible repayment period, account for 43.2 percent of total household debt. Furthermore, both installment loans and fixed-rate loans account for 65.7 percent of total household debt. Second, non-installment loans, whereby borrowers would bear greater monthly burden if interest rates were to rise, constitute only 34.3 percent of total household debt.

Chart 8 Households have less cushion against financial risks.
Ratio of debt to financial assets (DTFA)



Note: Calculations include only indebted households.
[1/] Households are classified into five quintiles based on their income per capita, with the 1st quintile having lowest monthly income per capita and the 5th quintile having highest monthly income per capita.
[2/] "Professional" households include managers, academicians and professionals, technicians, etc.
[3/] "Worker" households include those working in agriculture, forestry, fishery, machine control, clerkship, services, craftsmanship, manufacturing operation, etc.
Source: Socio-Economic Survey, National Statistical Office, calculated by Bank of Thailand

Chart 9 The ratio of household debt associated with floating rates increases slightly
Share of consumer loans classified by interest rate type



Note: *The fixed rate category includes "other loans that cannot be classified", which accounted for 4.83 percent as of 2018Q1.
*An increase in interest rate would affect borrowers of two types of loans:
- Installment loans (where a higher interest rate leads to a longer repayment period)
- Non-installment loans (where a higher interest rate affects monthly debt payments directly)

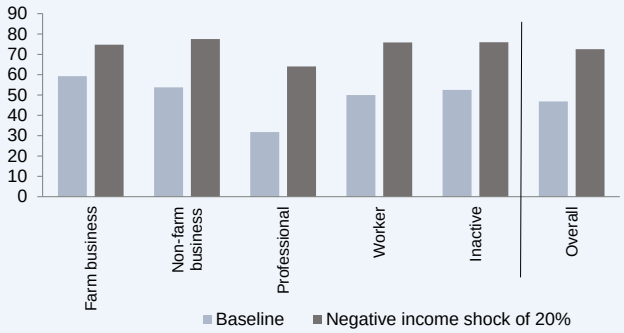
Source: Bank of Thailand

Regarding income shocks, indebted households are found to be liquidity constrained in order to meet debt repayments. Liquidity constraint here is defined as a situation when net household income, after deducting consumption and tax expenses, is below monthly debt burden. The proportion of household debt that is subject to such liquidity constraint is found to be as high as 46.8 percent of total household debt. In a stress test analysis, whereby household income falls by 20 percent and consumption is assumed to remain unchanged, the ratio of debt with liquidity constraint reaches 72.5 percent of total household debt, with all occupation groups suffering greater liquidity constraints (Chart 10).[18/] In the case of severe shocks, households might begin to adjust by reducing consumption and defaulting on their debt, and these could impact overall economic growth.

Chart 10 Indebted households may face liquidity constraints which affect their debt serviceability

Ratio of debt at risk[1/]

Percentage to total value of debt in each occupation



Note: [1/] Value of debt at risk refers to debt of households having incomes net of consumption and taxes that are insufficient to make full monthly debt payments assets to service their debts.

Source: Socio-Economic Survey, National Statistical Office; calculated by Bank of Thailand

In summary, although private consumption has been supported in part by a recent pickup in the pace of leveraging, such debt creation could undermine household financial positions. While the impact of higher interest rates on monthly debt burden is found to be limited, income shocks could put indebted households at risk. Therefore, if households as borrowers and financial institutions as lenders could change their behavior to become more financially disciplined, such endeavor could help reduce household debt as a share of GDP and consequently debt could become a driver of consumption without putting financial stability at risk in the future.

[18/] The ratio of debt with liquidity constraints may be lower than the abovementioned figure due to reporting error of the data collected in the socio-economic survey (SES).

3. Monetary Policy Decision

The Committee weighed various factors in order to maintain sustainable economic growth, while pursuing price stability and preserving financial stability.

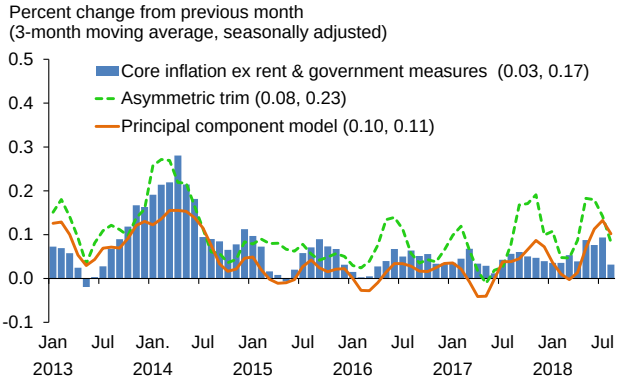
The Committee placed great emphasis on the strength and continuation of economic growth, development of headline inflation, and preserving financial stability. The Committee assessed that the economy would continue to gain further traction driven by strong momentum from both external and domestic demand, while downside risks to growth tilted downward but with a lesser degree. Inflation would rise in line with the previous assessment. However, risks to financial stability from the prolonged low interest rate must be monitored.

1. Economic growth. The Thai economy would continue to gain traction, driven by both external and domestic demand. However, merchandise exports of some products were affected by the U.S. trade protectionism measures in the recent period, but would be partially offset by benefits from the relocation of product base to Thailand for some industries in the period ahead. Meanwhile, the impact of the Phuket boat incident on the number of Chinese tourists was expected to be short-lived. With regard to domestic demand, household income, employment outside agricultural sector, as well as a continued improvement in consumer confidence would help boost private consumption going forward. However, structural changes in the labor market limited wage increases and household debt remained elevated. As a result, purchasing power would gradually recovering. Nevertheless, private investment growth would be supported by improvements in business confidence and exports, as well as investment in the EEC and the PPP projects where the prospects became more certain However, the progress on such investment projects in the period ahead must be monitored. Public expenditure would drive economic growth to the lesser extent due to limited efficiency in budget disbursement and a smaller expansion of compensation of civil servants given the policy to replace vacant job positions with contract workers. Nevertheless, risks to the growth projection were expected to tilt downward but with a smaller degree than the previous assessment. The Thai economy would still face risks from the external front that warranted close monitoring going forward. Such risks included the U.S. trade protectionism measures that could be additionally announced, particularly additional tariffs on imported cars and auto parts, retaliatory measures from major economies, as well as geopolitical risks that might affect economies of Thailand's trading partners.

2. Inflation development. Average headline inflation was projected to be within target in 2018 and 2019. The Committee viewed that inflation would rise in line with the previous assessment despite downside risks from volatile fresh food prices owing to weather conditions and agricultural output. Meanwhile, core inflation was expected to rise slowly as demand-pull price pressures would improve gradually (Chart 3.1). Nevertheless, the Committee viewed that there were various factors that led to a lower level of inflation than in the past and contributed to inflation that became less responsive to economic expansion despite economic expansion around full potential. Such factors included, in particular, structural changes such as higher productivity that resulted in lower costs of production and an expansion of e-commerce that led to lower costs of production and intensified price competition, minimum wage increases that did not sufficiently yield impact to inflation, and government's pricing mechanism to control domestic energy prices. The Committee saw the need to study the

impact of such factors on inflation dynamics in further details, as this could have significant implications on determining the inflation target as well as inflation outlook in the future. Moreover, **inflation expectations** according to surveys from businesses and professional forecasters were well in line with the inflation target, although short-term inflation expectations by professional forecasters slightly declined (Chart 3.2).

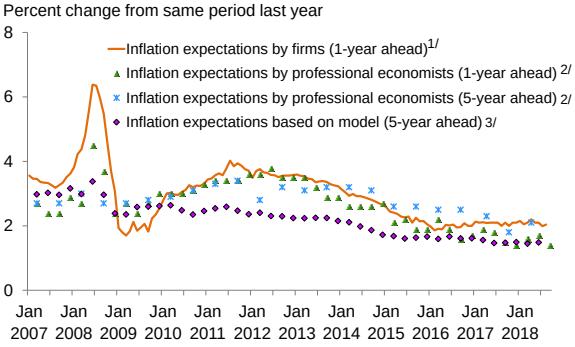
Chart 3.1 Demand-pull price pressures gradually increased.



Note: Data point indicated in () where the first value is %MoM (sa, 3mma) as of August 2018, while the second value is 2004 - 2014 average; Asymmetric trim excludes goods and services with most volatile price changes, removing the bottom 10 percentile and the top 6 percentile; Principal component model calculates changes in common statistical components that attribute price movements across categories of goods and services.

Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Chart 3.2 Short-term and long-term inflation expectations remained stable.



Source: 1/ Business Sentiment Survey of Bank of Thailand (BSI)
2/ Asia Pacific Consensus Forecast
3/ Calculations based on macro-finance term structure model with bond yield and macroeconomic data

3. The buildup of financial stability risks. The Committee viewed that financial stability remained sound overall, but there remained risks in certain pockets which might result in a buildup of vulnerabilities to overall financial stability in the future. Such risks included, first, **signs of increased vulnerability in the property sector**, although an oversupply of new condominiums showed somewhat improving signs as some developers postponed launches of their new projects, but there was still an oversupply of new condominiums in certain areas as well as existing ones waiting to be sold. Moreover, **competition among financial institutions in extending mortgage loans could lead to an increased vulnerability to financial stability** as reflected in (1) a rising share of new mortgage loans with loan-to-value (LTV) ratio exceeding 90 percent, (2) a rising share of new mortgage loans with increasing loan-to-income (LTI) ratio, and (3) an increasing share of mortgage loans extended to borrowers for a second or more homes. Such factors partly resulted in deteriorating credit standards as reflected in increased share of NPLs. The second financial stability risk was **household debt accumulation which had yet to show any clear signs of deleveraging, while debt serviceability of households and small businesses did not yet improve.** This was reflected in the household debt-to-GDP ratio which slowly decreased and remained at an elevated level, whereas low interest returns might affect the saving base in the future. Moreover, debt serviceability of SMEs with small credit lines and weak financial positions did not yet improve as reflected in NPLs of SMEs that remained high which might pose risks to growth in the long run. The third financial stability risk was the **continued search-for-yield behavior given a prolonged low interest rate environment that could lead to increased underpricing of risks.** These risks included saving cooperatives that continued to provide high returns to members resulting in a continued high growth of their assets, which in turn could pressure them to search for higher returns such as higher investment in bonds. In addition, some large saving cooperatives accelerated their funding through commercial bank

loans, while increasingly engaged in extending loans to other saving cooperatives. Consequently, this would increase systematic risks both within saving cooperatives and to the overall Thai financial system. Furthermore, given the prolonged low interest rate environment, the issuance of corporate bonds was largely concentrated among large businesses, which tended to invest more in non-core businesses and overseas enterprises. This would pose greater risks to business operations.

Nevertheless, the Committee viewed that certain risks to financial stability might be self-correctable through market mechanism. Others might not be corrected through market mechanism, and thus macroprudential measures could be applied to address and prevent certain risks in some economic sectors. Meanwhile, some Committee members were concerned about financial stability risks as vulnerabilities to the financial system started to become widespread and did not contain in their own pockets, partly owing to a prolonged low interest rate environment. Consequently, relying only on macroprudential measures to address those risks would not be sufficiently effective.

The Committee voted to keep the policy rate at 1.50 percent to maintain accommodative financial conditions which would support strong and continued economic growth as well as foster headline inflation to gradually rise and move within target in a sustainable manner, without accelerating the build-up of vulnerabilities to financial stability.

The Committee had to weigh various factors in formulating the most appropriate course of monetary policy and voted 6 to 1 at the meeting on August 8, 2018 and later 5 to 2 at the meeting on September 19, 2018 to maintain the policy rate at 1.50 percent. Monetary policy accommodation remained necessary to support a strong and continued growth of domestic demand and help foster headline inflation to gradually rise and move in line with the target. The Committee viewed that the current level of policy interest rate at 1.50 percent would facilitate sufficiently accommodative financial conditions as reflected in the new loan rate (NLR) which remained at a low level. Despite some increases, the real policy rate and government bond yields remained accommodative overall and continued to support business financing as reflected in a continued expansion in commercial bank loans extended to businesses as well as corporate bond issuances. However, some Committee members voted to raise the policy rate by 0.25 percentage point to 1.75 percent as economic expansion was sufficiently robust and expanded above its potential. In addition, monetary policy was exceptionally accommodative for a prolonged period, as reflected in the policy rate that was in the down cycle since 2011 and remained at a low level for the longest period compared with other past episodes. Therefore, a gradual reduction in the degree of monetary policy accommodation from the current level toward a normal level would not affect economic expansion. This would, in turn, help reduce risks to financial stability which would conducive to sustainable economic growth in the long run. Moreover, vulnerabilities in the financial system started to become widespread as an exceptionally accommodative financial conditions for a prolonged period induced households and businesses to underestimate potential changes to financial conditions. Hence, an increase in the policy interest rate deemed necessary in order to reduce risks to financial stability whose vulnerabilities started to accumulate and also to start building up policy space for the future.

In addition, as central banks of major economies and many regional countries started to reduce the degree of monetary policy accommodation, costs of funding in the global

financial market would likely increase and affect emerging markets (EMs) with fragile fundamentals first and other EMs afterward. Consequently, these would pose risks of highly volatile capital flows in the period ahead. In addition, the Committee viewed that monetary policy should particularly take into consideration benefits and costs associated with the context of the Thai economy. The reduction in the degree of monetary policy accommodation in the current juncture where the economy was facing significant uncertainties from both external and domestic fronts could affect growth momentum in the long run. Meanwhile, risks of abrupt capital outflow reversals driven by the interest rate differential between Thailand and other countries were limited, as Thailand's external stability remained sound which helped shore up investor confidence and cushion against volatilities in cross-border capital flows and in global financial markets. Meanwhile, risks to financial stability that were in part a result of the prolonged low interest rate environment remained manageable. Having considered policy trade-offs, the Committee viewed that monetary policy accommodation could be maintained at the current level.

Looking ahead, the Committee saw the need to maintain policy accommodation and discussed the conditions and appropriate timing to begin monetary policy normalization in the future. Should economic expansion continue to gain traction and inflation move in line with the target, the need for currently extra accommodative monetary policy would start to be gradually reduced and an increase in the policy rate to build up policy space in the future would be increasingly important. The Committee's evaluation of appropriate conditions would be data dependent, including careful assessment of the outlook of the economic growth and inflation, as well as risks especially on the external front.

4. Appendix

4.1 Table

Thai Economy Dashboard

Percent	2016	2017	2017			2018	
			Q2	Q3	Q4	Q1	Q2
GDP growth	3.3	3.9	3.9	4.3	4.0	4.9	4.6
Production							
Agriculture	-2.5	6.2	15.9	9.7	-1.3	6.5	10.4
Non-agriculture	3.8	3.7	3.0	4.0	4.7	4.8	4.1
Manufacturing	2.3	2.6	1.0	4.2	3.4	3.8	3.1
Construction	8.6	-2.3	-5.7	-1.6	-5.3	1.2	2.0
Wholesales and retail trade	5.3	6.3	6.0	6.4	6.9	7.0	7.2
Hotels and restaurants	9.9	8.5	7.0	6.9	15.3	12.8	9.4
Transport, storage, and communication	4.1	7.3	7.8	7.4	8.8	7.5	7.0
Financial intermediation	6.5	4.8	6.3	4.6	3.6	3.6	5.5
Real estate, renting, and business activities	3.2	4.6	4.2	4.7	5.8	4.9	3.2
Expenditure							
Domestic demand	2.8	2.1	1.8	2.5	2.0	3.3	3.7
Private consumption	3.0	3.2	2.9	3.4	3.4	3.7	4.5
Private investment	0.5	1.7	3.0	2.5	2.4	3.1	3.2
Government consumption	2.2	0.5	0.4	1.8	0.2	1.9	1.4
Public investment	9.5	-1.2	-6.9	-1.6	-6.0	4.0	4.9
Imports of goods and services	-1.0	6.8	7.2	6.5	7.5	8.7	7.5
imports of goods	-2.3	8.5	9.2	9.2	8.3	9.3	7.2
imports of services	4.6	-0.3	-1.0	-5.0	4.0	6.2	8.9
Exports of goods and services	2.8	5.5	5.1	6.9	7.4	6.0	6.4
exports of goods	0.3	5.6	4.9	8.2	6.6	4.7	7.4
exports of services	11.5	5.1	5.7	2.6	9.9	9.5	3.1
Trade balance (billion, U.S. dollars)	36.5	34.2	7.1	10.6	7.0	6.6	5.8
Current account (billion, U.S. dollars)	48.2	51.1	8.5	14.4	12.5	15.0	6.4
Financial account (billion, U.S. dollars)	-21.0	-19.1	-5.7	0.5	-6.9	-2.8	-7.1
International reserves (billion, U.S. dollars)	171.9	202.6	185.6	199.3	202.6	215.6	206.8
Unemployment rate (%)	1.0	1.2	1.2	1.2	1.1	1.2	1.1
Unemployment rate, seasonally-adjusted (%)	n.a.	n.a.	1.1	1.2	1.3	1.2	1.0

Source: Office of the National Economic and Social Development Board National Statistical Office and Bank of Thailand

Financial Stability Dashboard

Indicators	2016	2017	2017			2018			
			Q2	Q3	Q4	Q1	Q2	Jul	Aug
1. Financial market sector									
Bond market									
Bond spread (10 years - 2 years)	0.6	1.1	1.1	1.0	1.0	1.1	1.1	1.0	0.9
Equity market									
SET index (end of period)	1,542.9	1,753.7	1,574.7	1,673.2	1,753.7	1,776.3	1,595.6	1,701.8	1,721.6
Actual volatility of SET index[1/]	14.2	6.5	4.8	5.8	7.9	9.4	12.3	12.1	10.1
Price to Earnings ratio (P/E ratio) (times)	18.6	19.1	16.3	17.9	19.1	18.3	16.2	17.4	17.0
Exchange rate market									
Actual volatility of Thai baht (%annualized)[2/]	4.4	3.3	3.9	2.9	2.8	4.6	4.4	4.4	4.6
Nominal Effective Exchange Rate (NEER)	106.2	110.6	109.8	111.2	112.9	114.8	115.3	113.3	115.2
Real Effective Exchange Rate (REER)	100.6	103.6	102.8	104.2	105.7	106.3	107.1	105.1	106.8
2. Financial institution sector[3/]									
Minimum Lending Rate (MLR)[4/]	6.33	6.28	6.28	6.28	6.28	6.28	6.28	6.28	6.28
12-month fixed deposit rate[4/]	1.37	1.37	1.37	1.37	1.37	1.37	1.37	1.37	1.37
Capital adequacy									
Capital funds / Risk-weighted asset (%)	18.0	18.2	17.9	18.5	18.2	18.1	17.9	18.0	n.a.
Earning and profitability									
Net profit (billion, Thai baht)	198.5	187.1	48.8	46.5	40.6	50.2	56.5	n.a.	n.a.
Return on assets (ROA) (times)	1.2	1.1	1.2	1.1	1.1	1.1	1.2	n.a.	n.a.
Liquidity									
Loan to Deposit and B/E (%)	96.3	96.1	96.5	96.4	96.1	95.0	96.8	97.4	n.a.
3. Household sector									
Household debt to GDP (%)	79.3	78.0	78.0	77.8	78.0	77.6	n.a.	n.a.	n.a.
Financial assets to debt (times)	2.7	n.a.	2.8	2.7	2.6	n.a.	n.a.	n.a.	n.a.
Non-Performing Loans (NPLs) of commercial banks (%)									
Consumer loans	2.7	2.7	2.7	2.7	2.7	2.8	2.7	n.a.	n.a.
Housing loans	2.9	3.2	3.1	3.3	3.2	3.4	3.4	n.a.	n.a.
Auto leasing	1.8	1.6	1.7	1.6	1.6	1.5	1.5	n.a.	n.a.
Credit cards	3.7	2.6	3.4	2.8	2.6	3.2	2.4	n.a.	n.a.
Other personal loans	2.9	2.5	2.6	2.7	2.5	2.7	2.5	n.a.	n.a.
4. Non-financial corporate sector[5/]									
Operating profit margin (OPM) (%)	8.2	8.0	7.2	8.5	7.9	7.9	7.8	n.a.	n.a.
Debt to Equity ratio (D/E ratio) (times)	0.7	0.7	0.7	0.7	0.7	0.7	0.7	n.a.	n.a.
Interest coverage ratio (ICR) (times)	6.4	6.5	5.8	6.4	7.8	7.2	6.5	n.a.	n.a.
Current ratio (times)	1.6	1.7	1.6	1.6	1.7	1.7	1.7	n.a.	n.a.
Non-Performing Loans (NPLs) of commercial banks (%)									
Large businesses	1.5	1.8	1.8	1.7	1.8	1.7	1.7	n.a.	n.a.
SMEs	4.3	4.4	4.4	4.6	4.4	4.5	4.5	n.a.	n.a.

Note:

[1/] Calculated by 'annualized standard deviation of return' method

[2/] Daily volatility (using exponentially weighted moving average method)

[3/] Based on data of all commercial banks

[4/] Average value of 5 largest Thai commercial banks

[5/] Only listed companies on Stock Exchange of Thailand (median value); with data revisions

Financial Stability Dashboard (continue)

Indicators	2016	2017	2017			2018			
			Q2	Q3	Q4	Q1	Q2	Jul	Aug
5. Real estate sector									
Number of approved mortgages from commercial banks (Bangkok and Vicinity) (units)									
Total	61,452	62,665	15,086	16,859	18,476	12,856	15,230	5,934	n.a.
Single-detached and semi-detached houses	13,409	13,907	3,544	3,774	3,787	3,162	3,365	1,201	n.a.
Townhouses and commercial buildings	20,187	20,536	4,947	5,604	5,670	4,248	4,922	1,962	n.a.
Condominiums	27,856	28,222	6,595	7,481	9,019	5,446	6,943	2,771	n.a.
Number of new housing units launched for sale (Bangkok and Vicinity) (units)									
Total	110,575	114,477	25,256	35,434	24,863	25,026	18,615	8,705	10,690
Single-detached and semi-detached houses	19,433	14,280	2,830	4,874	3,608	3,758	2,629	536	1,691
Townhouses and commercial buildings	32,792	36,571	7,665	9,831	7,892	6,032	6,348	1,845	2,504
Condominiums	58,350	63,626	14,761	20,729	13,363	15,236	9,638	6,324	6,495
Housing price index (2009 = 100)									
Single-detached houses (including land)	130.8	130.9	129.6	131.6	133.9	137.7	137.8	138.3	n.a.
Townhouses (including land)	137.6	141.2	140.0	142.6	143.7	145.9	149.5	150.5	n.a.
Condominiums	166.2	171.0	168.8	169.8	175.4	182.3	176.7	178.0	n.a.
Land	171.2	171.7	164.2	172.9	178.3	175.7	177.2	177.3	n.a.
6. Fiscal sector									
Public debt to GDP (%)	40.8	41.2	41.3	41.9	41.2	41.2	41.0	40.9	n.a.
7. External sector									
Current account balance to GDP (%)[6/]	11.7	11.2	7.8	12.5	10.2	11.6	5.2	n.a.	n.a.
External debt to GDP (%)[7/]	32.5	35.3	34.0	35.5	35.3	35.4	33.4	n.a.	n.a.
External debt (billion, U.S. dollars)	132.2	149.4	140.3	148.1	149.4	152.7	146.8	145.4	n.a.
Short-term (%)	41.2	42.1	39.4	40.9	42.1	41.2	41.2	40.7	n.a.
Long-term (%)	58.8	57.9	60.6	59.1	57.9	58.8	58.8	59.3	n.a.
International reserves / Short-term external debt (times)	3.2	3.2	3.4	3.3	3.2	3.4	3.4	3.5	n.a.

Note:

[6/] Current account / Nominal GDP at the same quarter

[7/] External debt / 3-year average nominal GDP

Table: Probability distribution of GDP growth forecast

Percent	2018		2019				2020	
	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
> 9	0	0	0	0	2	4	4	5
8-9	0	0	0	1	4	5	5	5
7-8	0	1	2	3	7	9	8	8
6-7	2	5	5	7	12	13	12	11
5-6	12	15	12	12	16	16	15	14
4-5	32	25	19	17	17	16	15	14
3-4	34	25	22	19	16	14	14	13
2-3	17	17	19	16	12	10	11	11
1-2	4	8	12	12	7	6	8	8
0-1	0	2	6	7	4	3	5	5
(-1)-0	0	1	2	3	2	2	2	3
(-2)-(-1)	0	0	1	1	1	1	1	1
(-3)-(-2)	0	0	0	0	0	0	0	1
< (-3)	0	0	0	0	0	0	0	0

Table: Probability distribution of headline inflation forecast

Percent	2018		2019				2020	
	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
> 4.0	0	0	0	1	1	2	2	3
3.5-4.0	0	0	1	2	2	2	3	3
3.0-3.5	0	0	2	3	3	4	4	5
2.5-3.0	2	1	5	6	5	7	6	7
2.0-2.5	10	5	9	8	8	9	9	9
1.5-2.0	25	11	13	11	10	11	10	10
1.0-1.5	30	18	15	12	12	12	11	11
0.5-1.0	21	20	15	12	12	11	11	11
0.0-0.5	9	18	13	11	11	10	10	10
(-0.5)-0.0	3	13	10	10	10	9	9	9
(-1.0)-(0.5)	1	7	7	8	8	7	7	7
(-1.5)-(1.0)	0	4	5	6	6	5	6	6
(-2.0)-(-1.5)	0	1	3	4	5	4	4	4
< -2.0	0	1	2	6	7	6	7	7

Table: Probability distribution of core inflation forecast

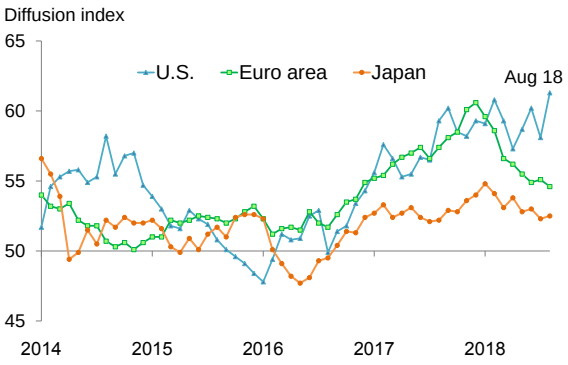
Percent	2018		2019				2020	
	Q3	Q4	Q1	Q2	Q3	Q4	Q1	Q2
> 4.0	0	0	0	1	1	2	2	3
3.5-4.0	0	0	1	2	2	2	2	3
3.0-3.5	0	0	2	3	3	4	4	4
2.5-3.0	2	2	5	6	5	6	6	6
2.0-2.5	10	5	9	8	7	9	8	8
1.5-2.0	25	12	13	11	10	10	10	10
1.0-1.5	30	18	15	12	11	12	11	11
0.5-1.0	21	20	15	12	12	12	11	11
0.0-0.5	9	18	13	11	11	11	10	10
(-0.5)-0.0	3	12	11	10	10	9	9	9
(-1.0)-(0.5)	0	7	7	8	8	7	7	7
(-1.5)-(1.0)	0	3	5	6	6	6	6	6
(-2.0)-(-1.5)	0	1	3	4	5	4	4	4
< -2.0	0	1	2	6	7	6	7	7

4.2 Data pack

Global Economy

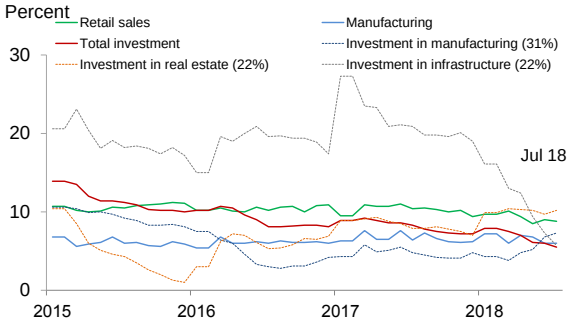
Overall Thailand's trading partner economies continued to expand and would remain a key driver of Thai exports going forward. However, some slowdown was expected due to the impact of US trade protectionism policy and intensifying retaliatory measures from major economies following the 200 billion U.S. dollar tariffs imposed on Chinese imported goods. Headline inflation rose due mainly to oil prices.

Manufacturing Purchasing Manager Index



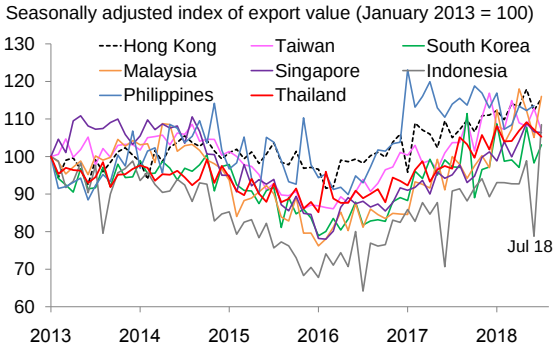
Sources: Bloomberg and Eurostat

China's economic indicators (Change from same period last year)



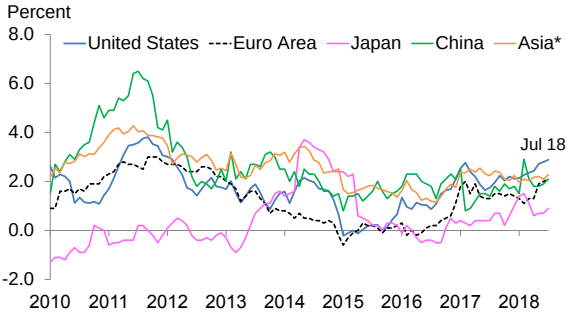
Note: ( ) denotes share to total investment
Source: CEIC

Asian exports



Source: CEIC

Inflation of Thailand's major trading partners

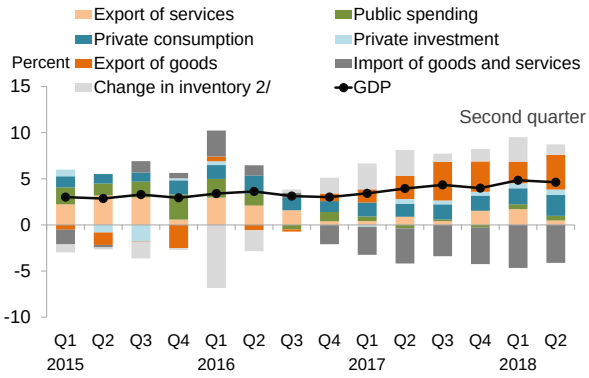


Note: * Average of headline inflation in Indonesia, South Korea, Malaysia, the Philippines, Singapore and Taiwan
Source: CEIC

Thai Economy

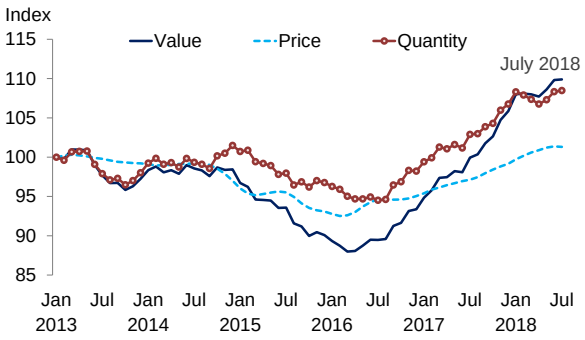
Economic growth continued to gain further fraction, underpinned by exports which expanded across almost all major markets and product categories. Exports of services continued to expand despite some slowdown after the Phuket tour boat sinking incident. Private consumption increased driven by improvements in employment and income. Private investment also increased, owing in part to the clarity of EEC and PPP projects which helped boost investor confidence. Public expenditure continued to drive the economy despite somewhat lower-than-expected growth.

Contribution to Thailand's GDP growth[1/]



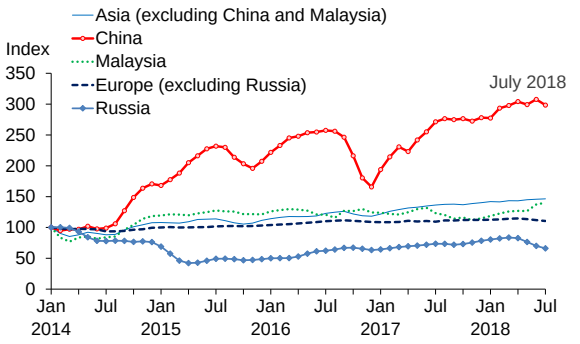
Note: [1/] Calculated by Chain Volume Measure method (CVM)
[2/] Change in inventory and statistical discrepancy
Source: Office of National Economic and Social Development Board, calculations by Bank of Thailand

Thai exports (excluding gold): value, price and quantity
(3-month moving average, seasonally adjusted; January 2013 = 100)



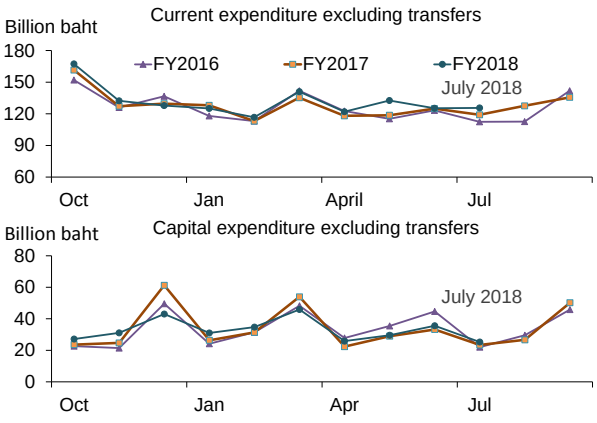
Source: Customs Department and Ministry of Commerce, calculations by Bank of Thailand

Index of foreign tourists classified by nationality
(three-month moving average, seasonally adjusted; January 2014 = 100)



Source: Department of Tourism

Public spending by central government

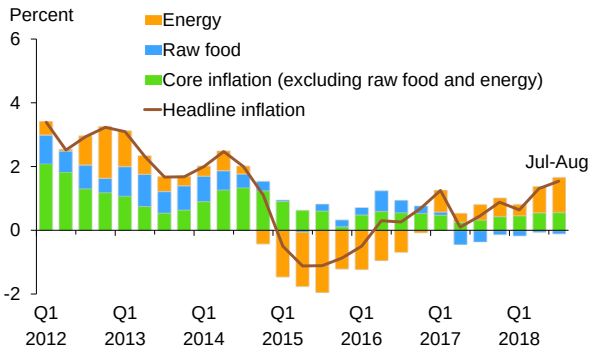


Source: Bureau of Budget, Fiscal Policy Office

Inflation

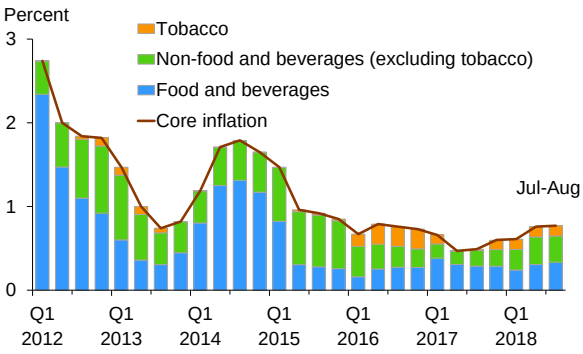
Headline inflation rose due mainly to energy prices. Core inflation remained stable with prices of most goods and services slowly trended up. The short-term (one-year-ahead) inflation expectations according to the survey of businesses remained largely unchanged, while those of professional forecasters slightly lowered. Meanwhile, long-term (five-year-ahead) inflation expectations according to a survey of professional forecasters in April 2018 increased in comparison to the previous survey in October 2017.

Contribution to headline inflation



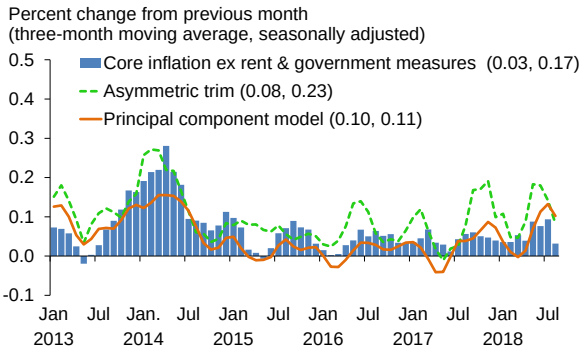
Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Contribution to core inflation



Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

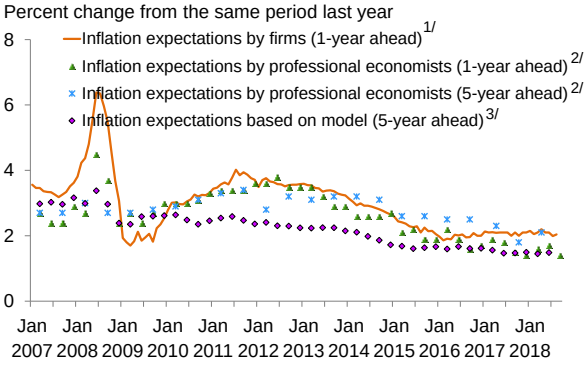
Underlying inflation indicators



Note: Data point indicated in () where the first value is %MoM (sa, 3mma) as of August 2018, while the second value is 2004 - 2014 average; **Asymmetric trim** excludes goods and services with most volatile price changes, removing the bottom 10 percentile and the top 6 percentile; **Principal component model** calculates changes in common statistical components that attribute price movements across categories of goods and services.

Source: Bureau of Trade and Economic Indices, Ministry of Commerce, calculations by Bank of Thailand

Inflation expectations

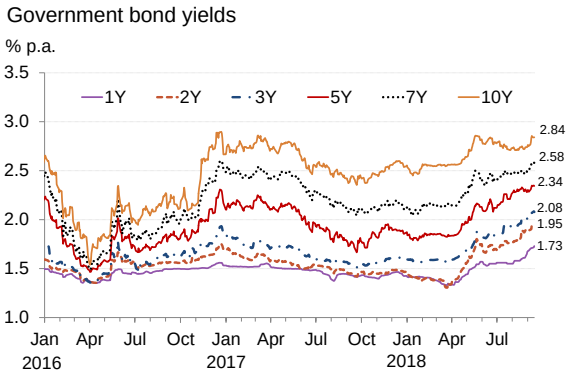


Source: 1/ Business Sentiment Survey of Bank of Thailand (BSI)
2/ Asia Pacific Consensus Forecast
3/ Calculations based on macro-finance term structure model with bond yield and macroeconomic data

Financial Conditions

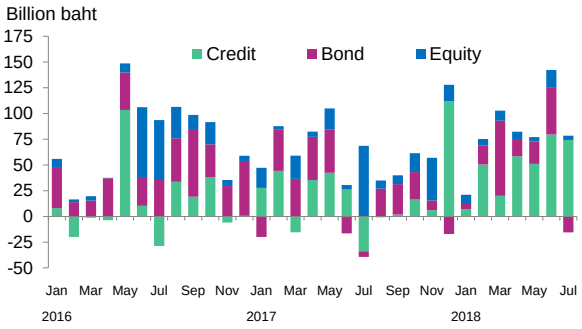
Short-term government bond yields remained below the policy rate. Overall, government bond yields increased, especially medium-term yields due to better-than-expected outturns of Thailand's latest economic outturns. Long-term yields, however, increased only slightly due to demand from foreign investors. Private credit accelerated for both corporates and households, in line with the continuation of economic growth. The baht appreciated against the U.S. dollar due to Thailand's better-than-expected economic outturns, expectations that the policy rate would be raised sooner than previous assessment, and the return of investors to emerging market economies with sound external stability that included Thailand, South Korea, and Taiwan.

Movements in medium- and long-term government bond yields were driven mostly by external factors



Source: Thai Bond Market Association (Thai BMA) (data as of 18 September 2018)

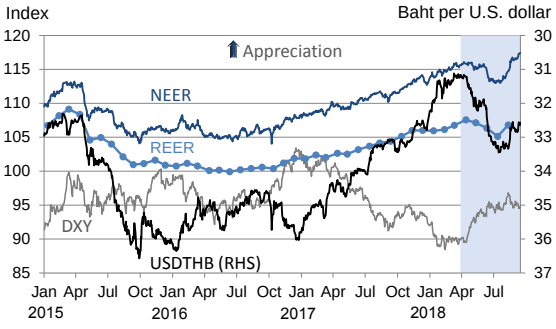
Total corporate financing by instrument*



Note: * Monthly change in outstanding of corporate loans (seasonally adjusted), corporate bonds excluding commercial banks, and newly issued equities.

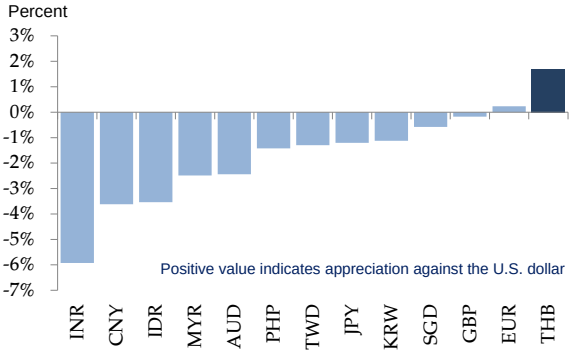
Sources: Bank of Thailand and Thai Bond Market Association (Thai BMA)

The Thai baht vis-a-vis U.S. dollar (USDTHB), Nominal Effective Exchange Rate (NEER), and the Dollar Index (DXY)



Source: Bank of Thailand and Reuters (data as of 18 September 2018)

Currency movements vis-a-vis the U.S. dollar (18 Sep 2018 compared to 29 Jun 18)

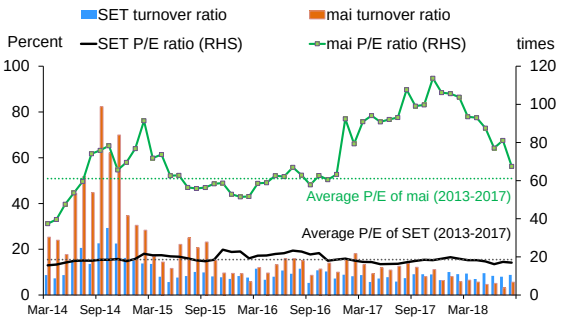


Source: Bank of Thailand and Reuters (data as of 18 September 2018)

Stability: financial markets

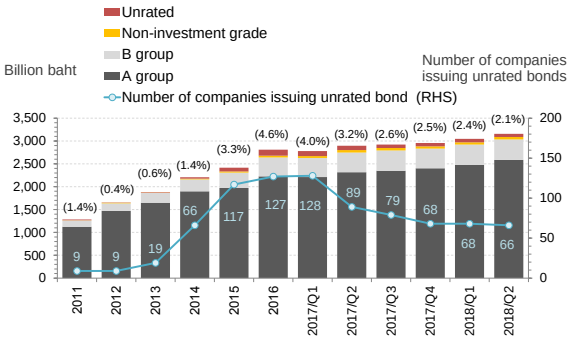
The price-to-earning (P/E) ratio of the Stock Exchange of Thailand (SET) stayed slightly below the historical average, while that of the Market for Alternative Investment (mai) continued to trend down albeit above the historical average. The share of unrated bonds issuance fell to 2.1 percent of total corporate bond outstanding in the second quarter of 2018.

Current price-to-earning ratio and turnover ratio of SET and mai



Source: Stock Exchange of Thailand (as of Sep 2018)

Corporate bonds outstanding

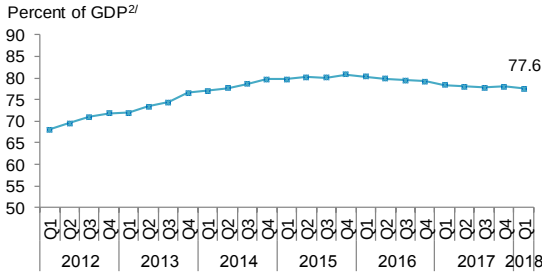


Note: ( ) represents percent of unrated bonds in total corporate bonds
Source: Thai Bond Market Association (Thai BMA)

Stability: household sector

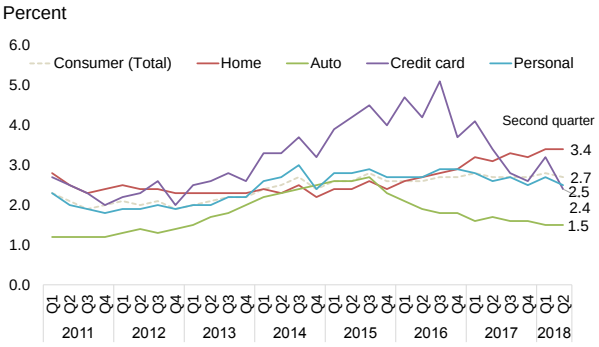
The ratio of household debt to GDP remained at a high level despite a slight decline from the previous quarter. As of the end of the second quarter of 2018, credit quality of consumer loans deteriorated, partly due to debt restructuring and write-offs. Meanwhile, mortgage loan quality continued to worsen. Thus, household debt serviceability would warrant monitoring going forward.

Household debt[1/]



Note: [1/] Loans to households by financial institutions
[2/] Calculated by averaging the 4 latest quarterly GDP
[3/] Household debt and GDP data are revised. This results in the different debt to GDP ratios compared to the last MPR.
Source: Bank of Thailand

Share of non-performing loans (NPL) in consumer loans, classified by loan type

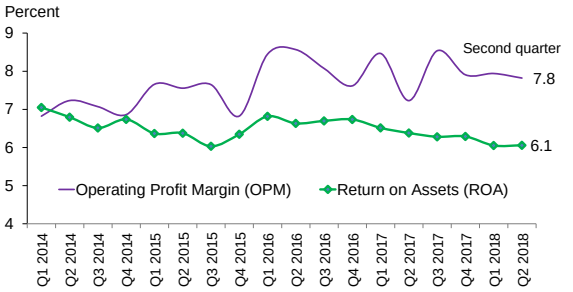


Source: Bank of Thailand

Stability: corporate sector

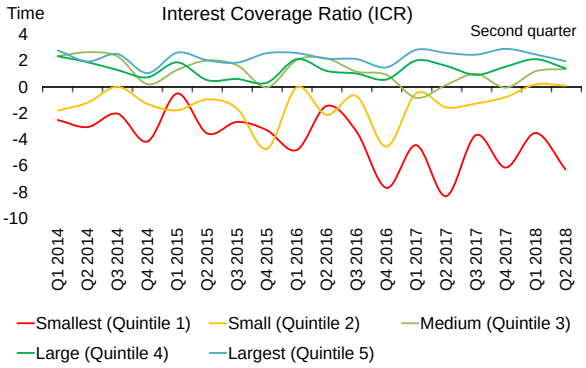
In the second quarter of 2018, stability of the corporate sector remained sound in line with continued economic growth. The ratio of corporate debt to GDP continued to decline and financial positions, especially for large corporates, improved. However, small enterprises were still subject to vulnerable financial positions, partly because their competitiveness deteriorated. Debt serviceability of small enterprises and businesses in some sectors, such as construction and commerce, would warrant monitoring.

Operating Profit Margin (OPM) and Return on Assets (ROA)*



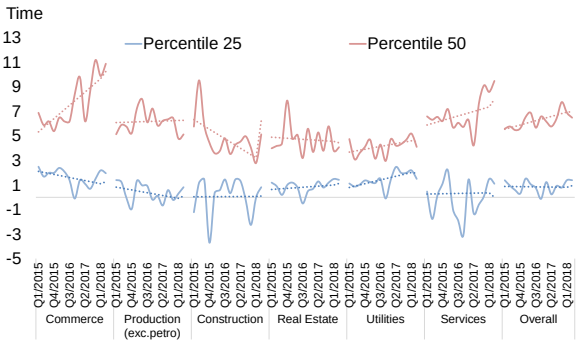
Note: * Median estimates; ROA is returns to average assets.
OPM is operating profits to total sales.
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Debt serviceability at 25th percentile of each group of firm size



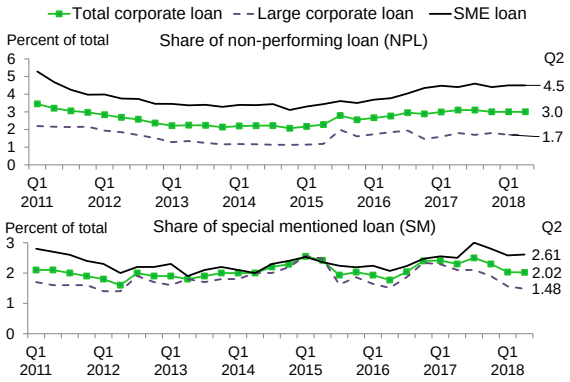
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Interest Coverage Ratio, classified by sectors



Note: * production exclude Petroleum and chemicals
Source: Stock Exchange of Thailand, calculation by Bank of Thailand

Loan quality of corporate sector

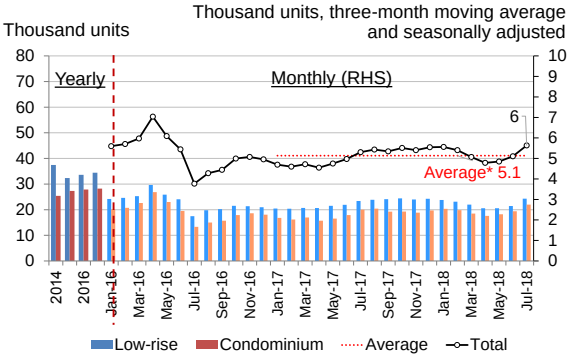


Source: Bank of Thailand

Stability: real estate

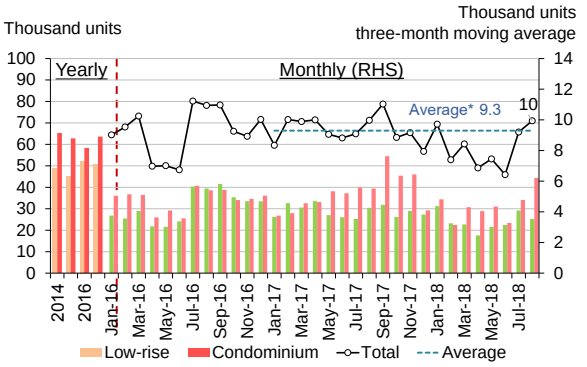
Demand for residences trended up, as reflected in the number of residential units with newly approved loans that continued to rise above the historical average. Demand rose for both low-rise and condominium residential units. The supply increased in line with increased newly opened residential units as planned by property developers since the beginning of the year. Property prices slightly increased due mainly to higher demand.

Residential units in Bangkok and its vicinity with approved mortgages by commercial banks



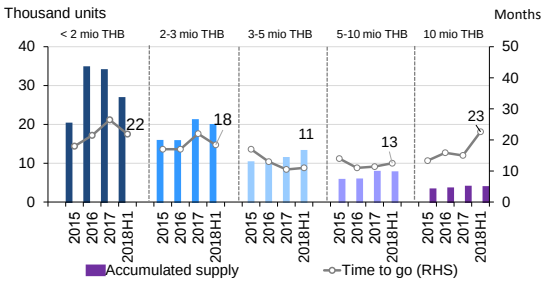
Note: *Average during 2014-2017
Source: Bank of Thailand

New residential projects launched in Bangkok and its vicinity



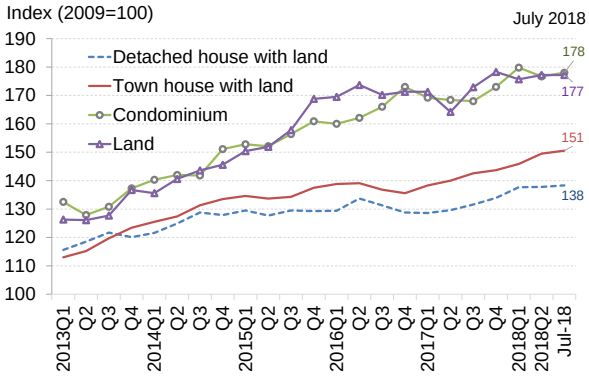
Note: *Average during 2014-2017
Source: Agency for Real Estate Affairs (AREA), calculation by Bank of Thailand

Condominium inventory in Bangkok and vicinity and 'Time to go'



Note: 'Time to go' is the time taken for all real estate inventory to be sold out at the average sales rate per month (since projects launched) given no additional supply.
Source: AREA and calculation by Bank of Thailand

Real estate price indices

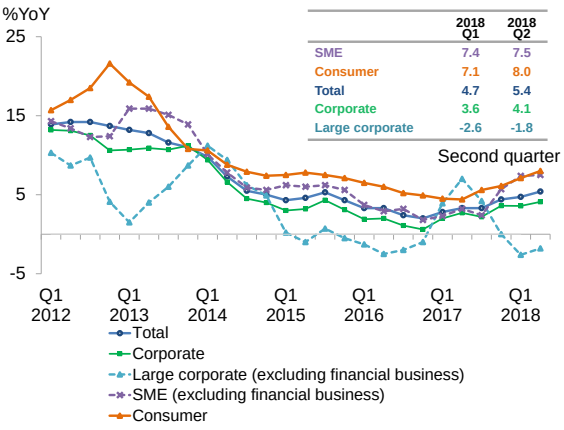


Source: Bank of Thailand

Stability: financial institutions

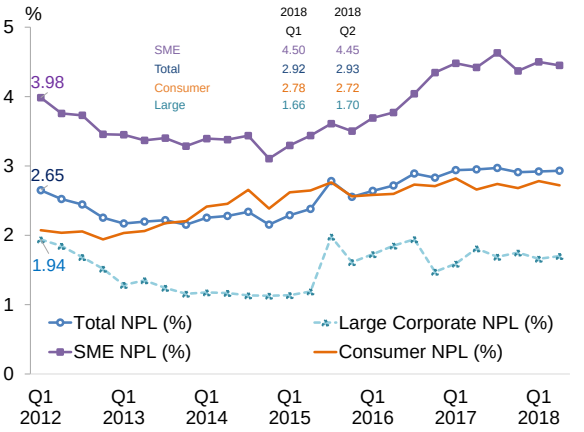
Financial institutions maintained strong financial positions, as reflected in high levels of capital buffers among commercial banks to cushion against risks should credit quality deteriorate. In the second quarter of 2018, commercial bank credits continued to grow for both business and consumer loans. The overall NPL ratio stabilized at a high level, especially for loans extended to SMEs and mortgage loans extended to households.

Credit growth in the commercial bank system



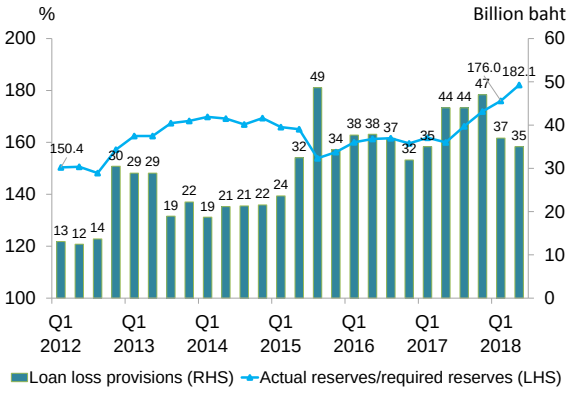
Source: Bank of Thailand

Non-performing loan (NPL)



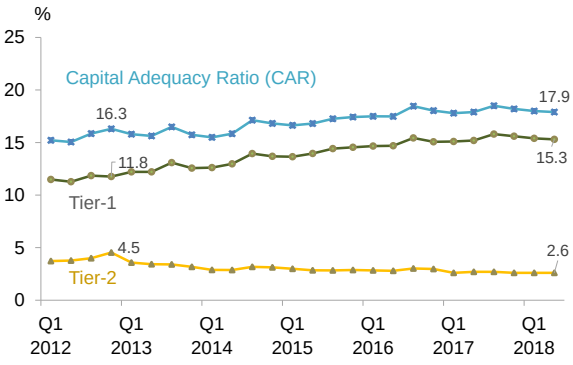
Source: Bank of Thailand

Provisions in commercial bank system



Source: Bank of Thailand

Capital buffers in commercial bank system

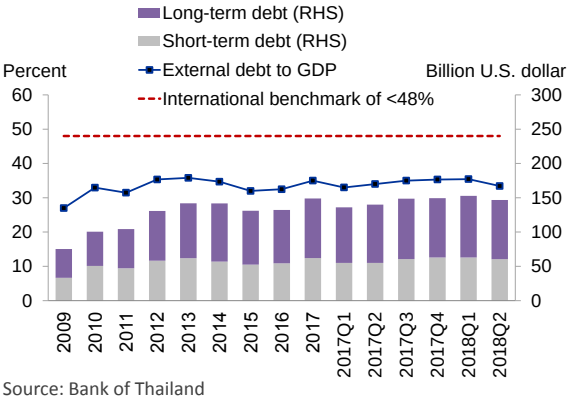


Source: Bank of Thailand

Stability: external position

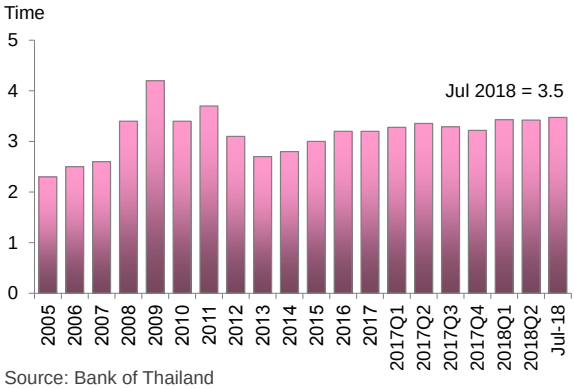
Thailand's external stability remained sound, as reflected in high levels of international reserves and a sustained current account surplus. Moreover, the ratio of external debt to GDP was below an international benchmark. This would help the Thai economy to be resilient against volatilities in global financial mar ket.

Thailand's external debt



Source: Bank of Thailand

Reserve to short-term debt

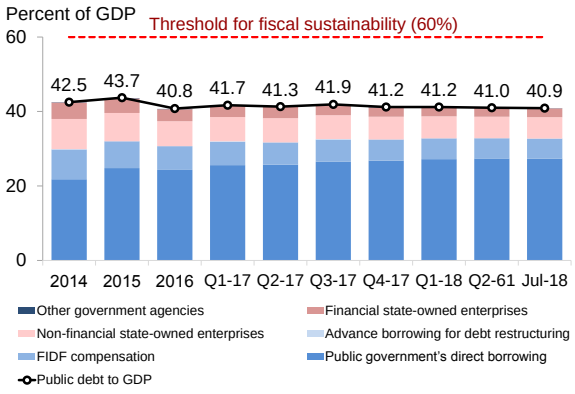


Source: Bank of Thailand

Stability: fiscal sector

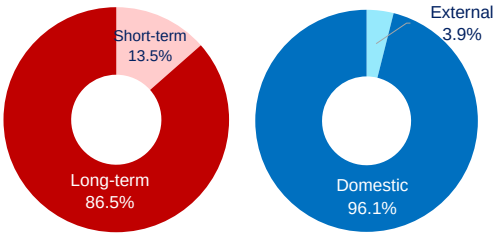
Fiscal stability remained sound. The ratio of public debt to GDP stayed below the sustainability threshold.

Public debt to GDP



Note: Calculated by GDP with Chain Volume Measure
Source: Public Debt Management Office

Outstanding debt as of July 2018



Note: Share of short-term and long-term debt calculated from remaining duration until maturity
Source: Public Debt Management Office



Pursuing Sustainable
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